Computer Vision —— Object Recognition

2026/9/22

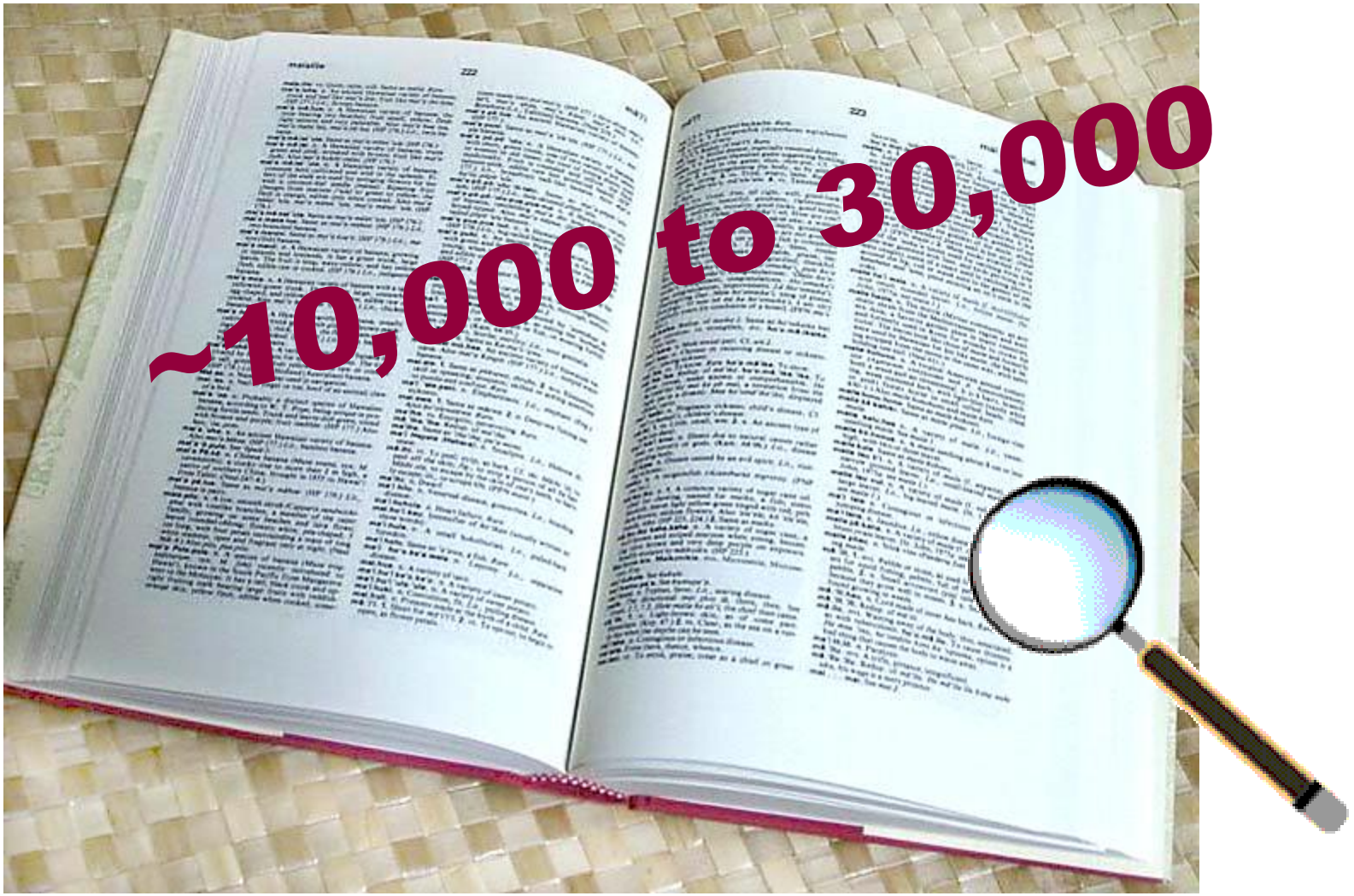
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Slides from CV course with Rob Fergus, Fei-Fei Li, and Antonio Torralba

Agenda

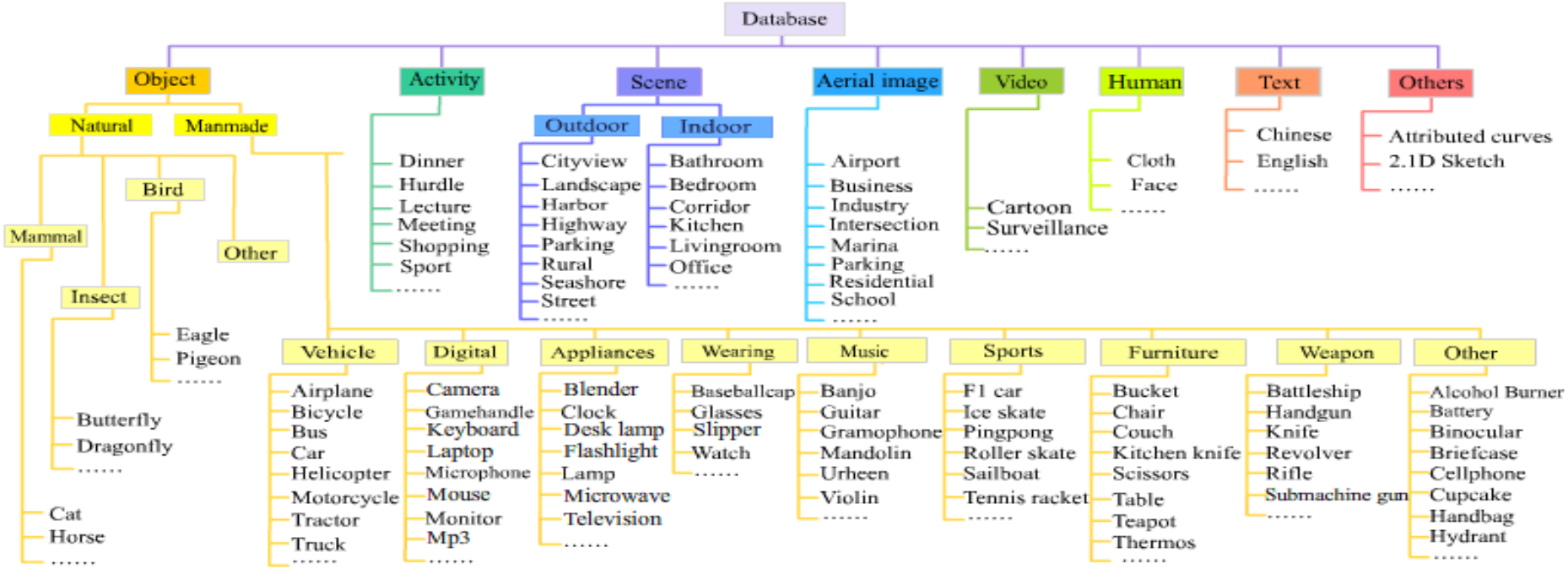
- Introduction
- Objects & Faces recognition
- Bag-of-words models
- Part-based models

How many object categories are there?



Biederman 1987

How many object categories are there?



细粒度目标识别
fine-grained classification

Biederman 1987

So what does object recognition involve?



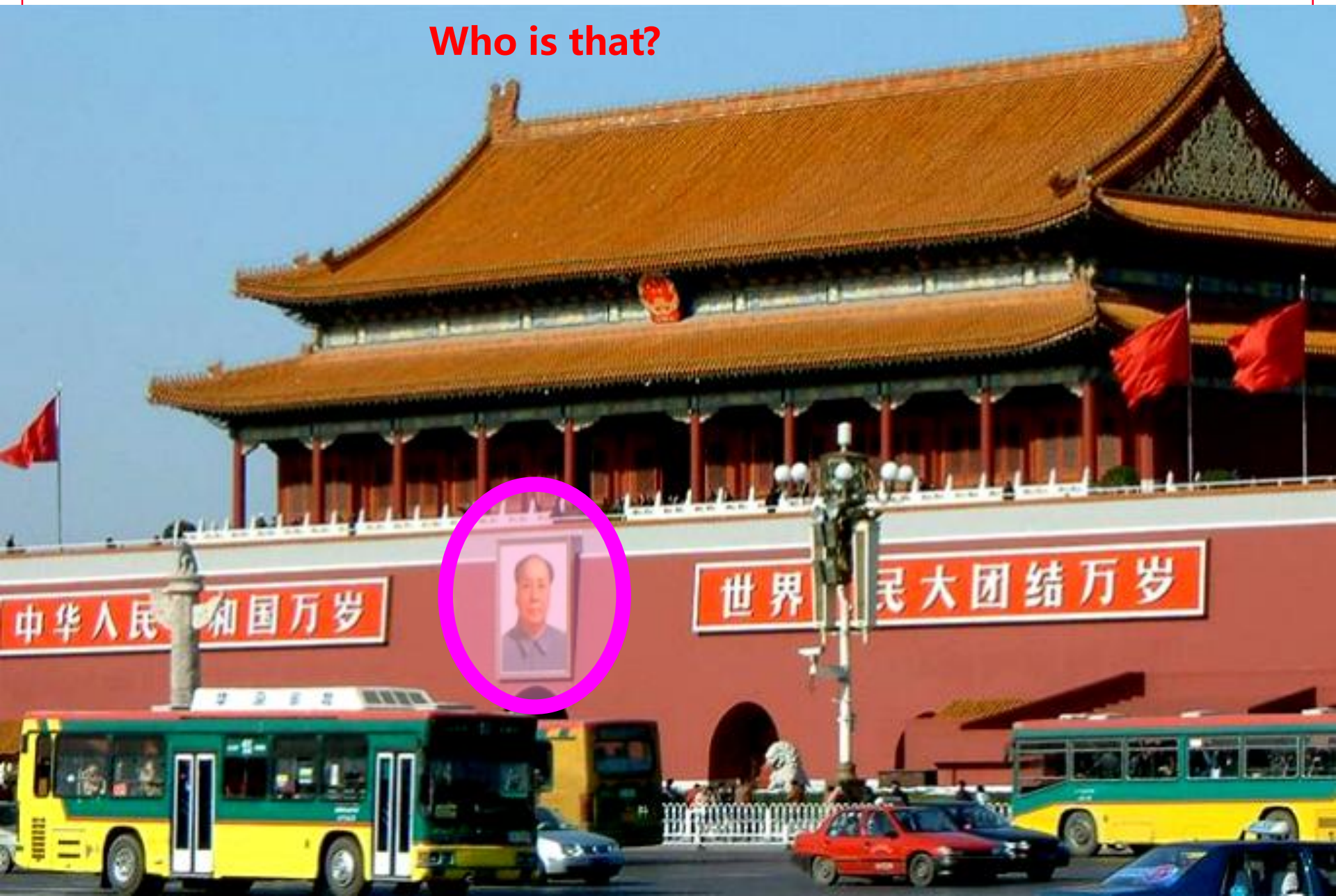
Verification: is that a bus?



Detection: where are cars?



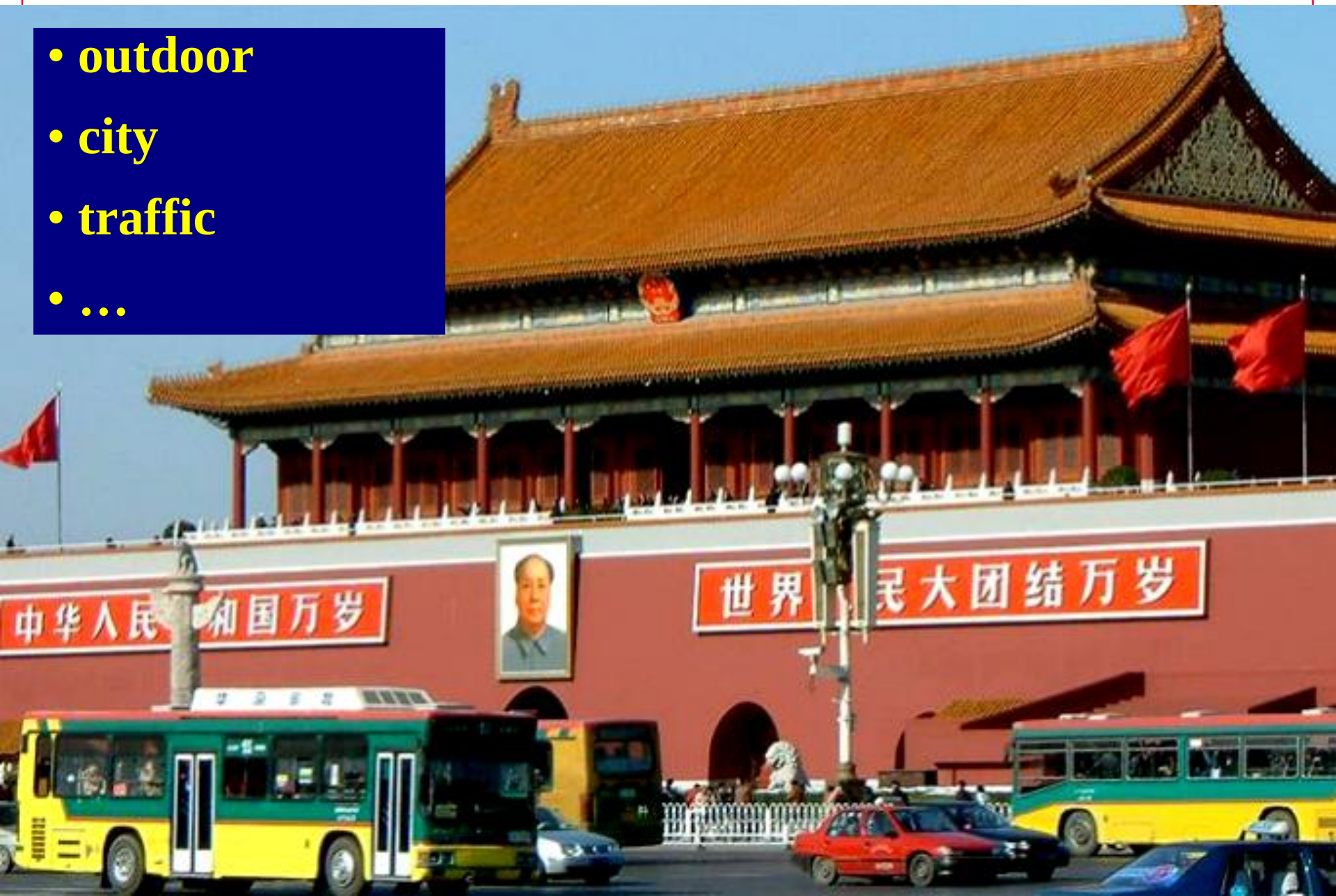
Identification: is that a picture of Mao?



Object categorization

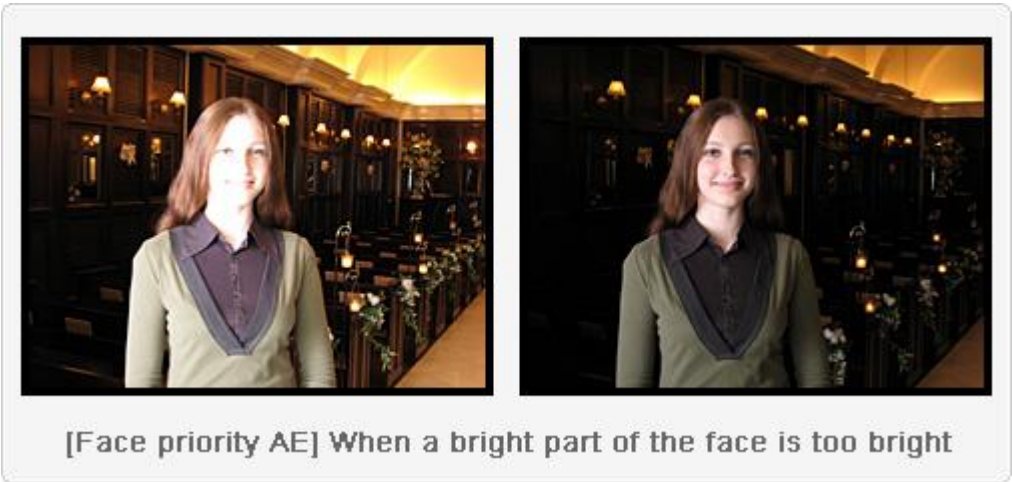


Scene and context categorization



Applications: Photography





Google driverless car





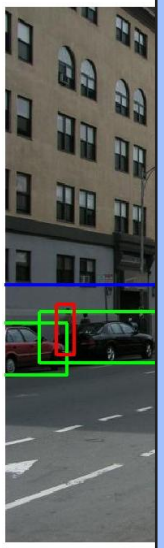
Google driverless car

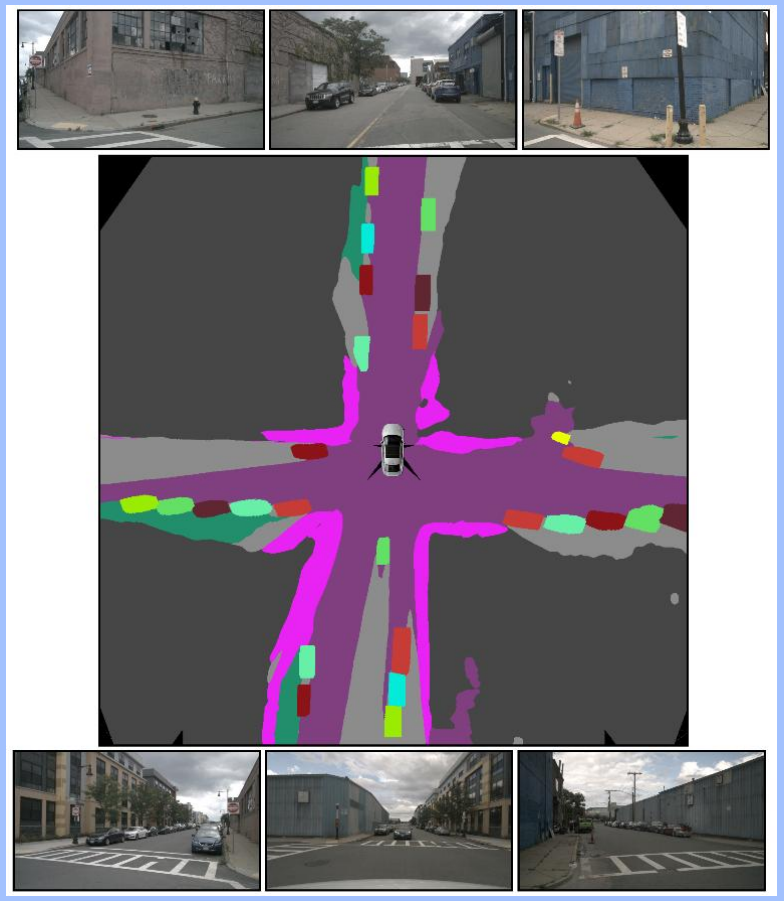




Application: Assisted driving

Pedest

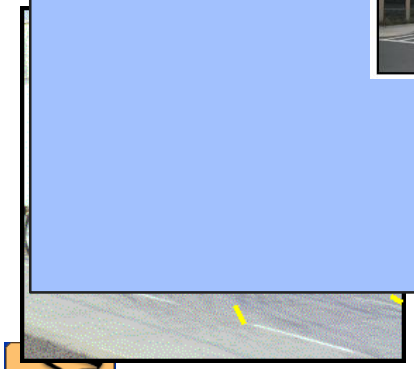






Lane d

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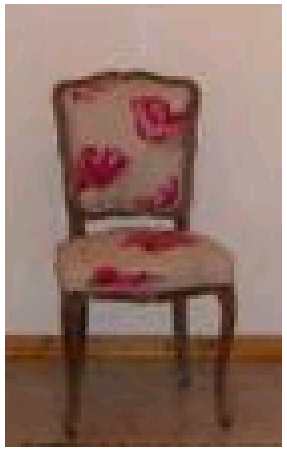


systems

点云鸟瞰图
BEV（Bird’s-eye-view)

Object recognition
Is it really so hard?

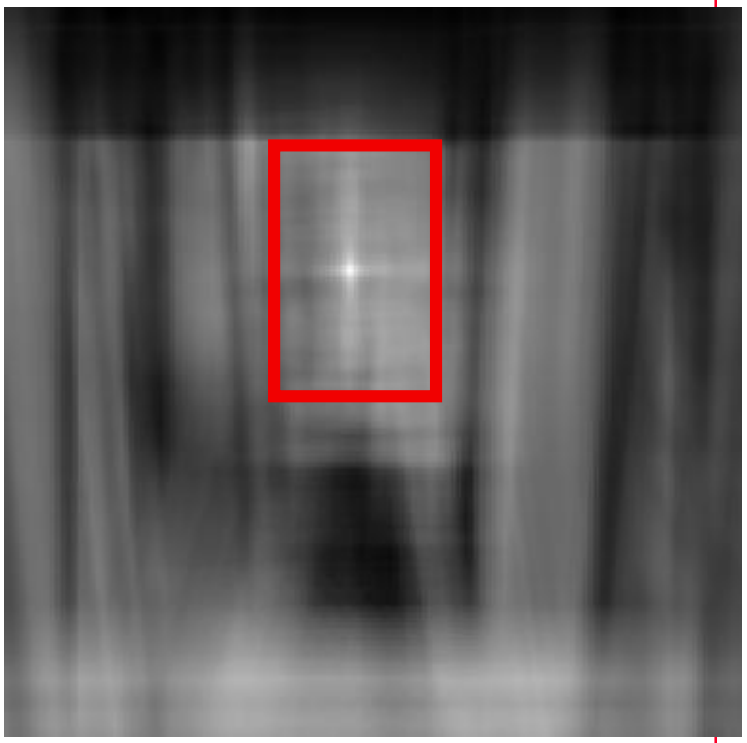
This is a chair



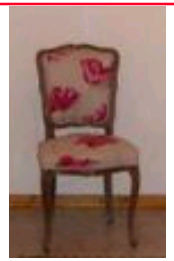
Find the chair in this image



Output of normalized correlation

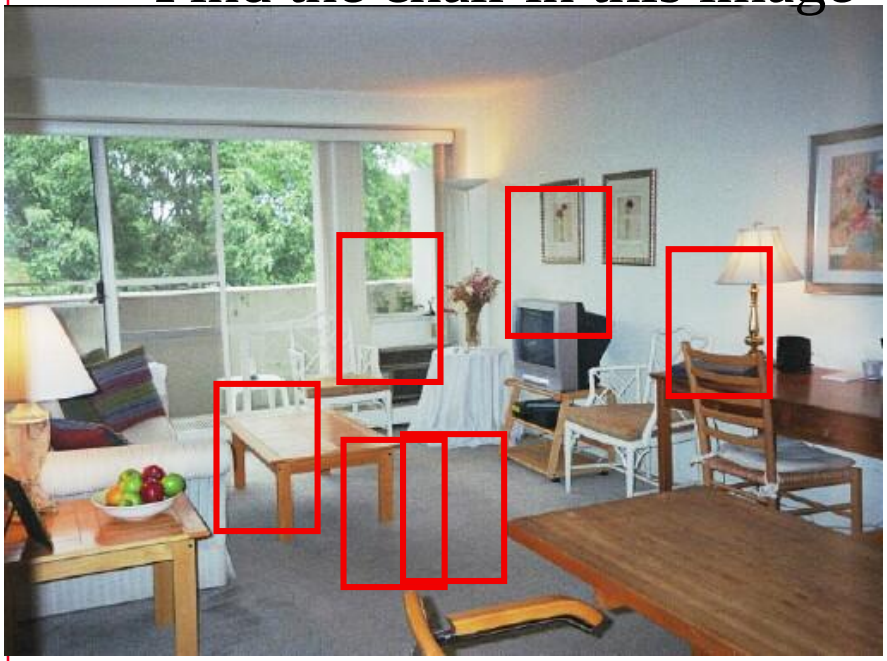


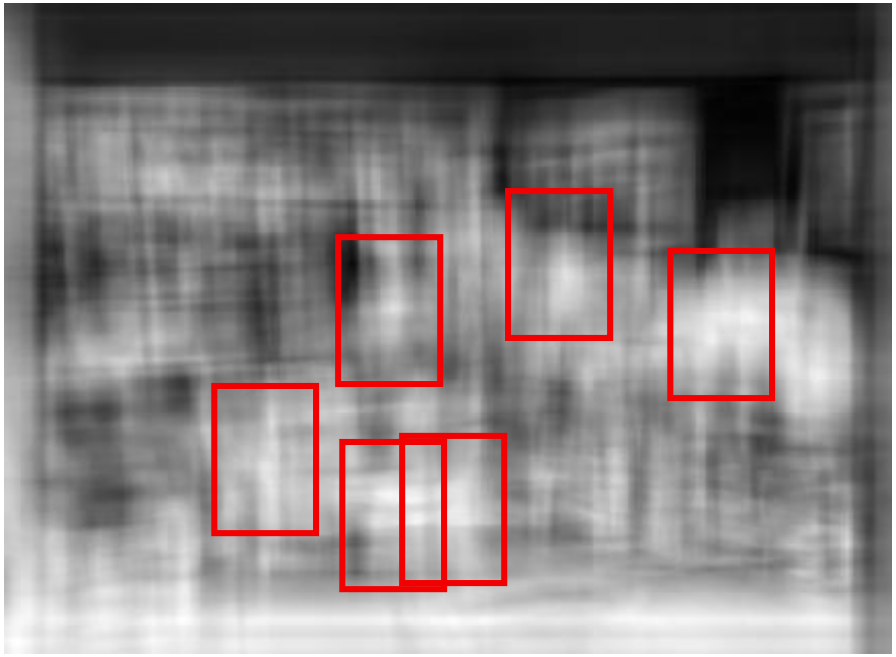
Slide: A. Torralba



Object recognition
Is it really so hard?

Find the chair in this image





Pretty much garbage
Simple template matching is not going to make it

A "popular method is that of template matching, by point to point correlation of a model pattern with the image pattern. These techniques are inadequate for three-dimensional scene analysis for many reasons, such as occlusion, changes in viewing angle, and articulation of parts." Nivatia & Binford, 1977.

Slide: A. Torralba

Challenges 1: view point variation







Michelangelo 1475-1564

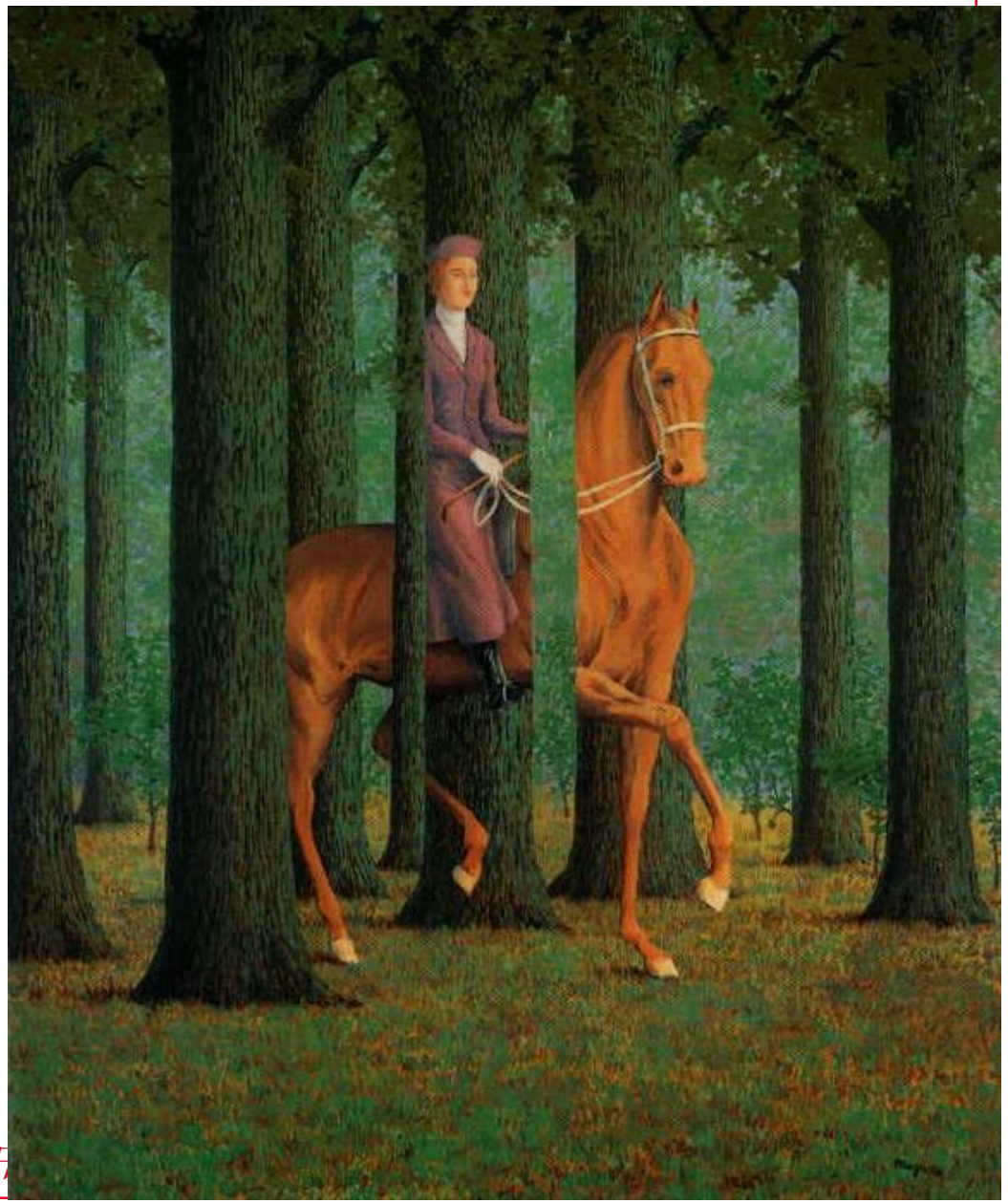
Challenges 2: illumination





slide credit: S. Ullman

Challenges 3: occlusion

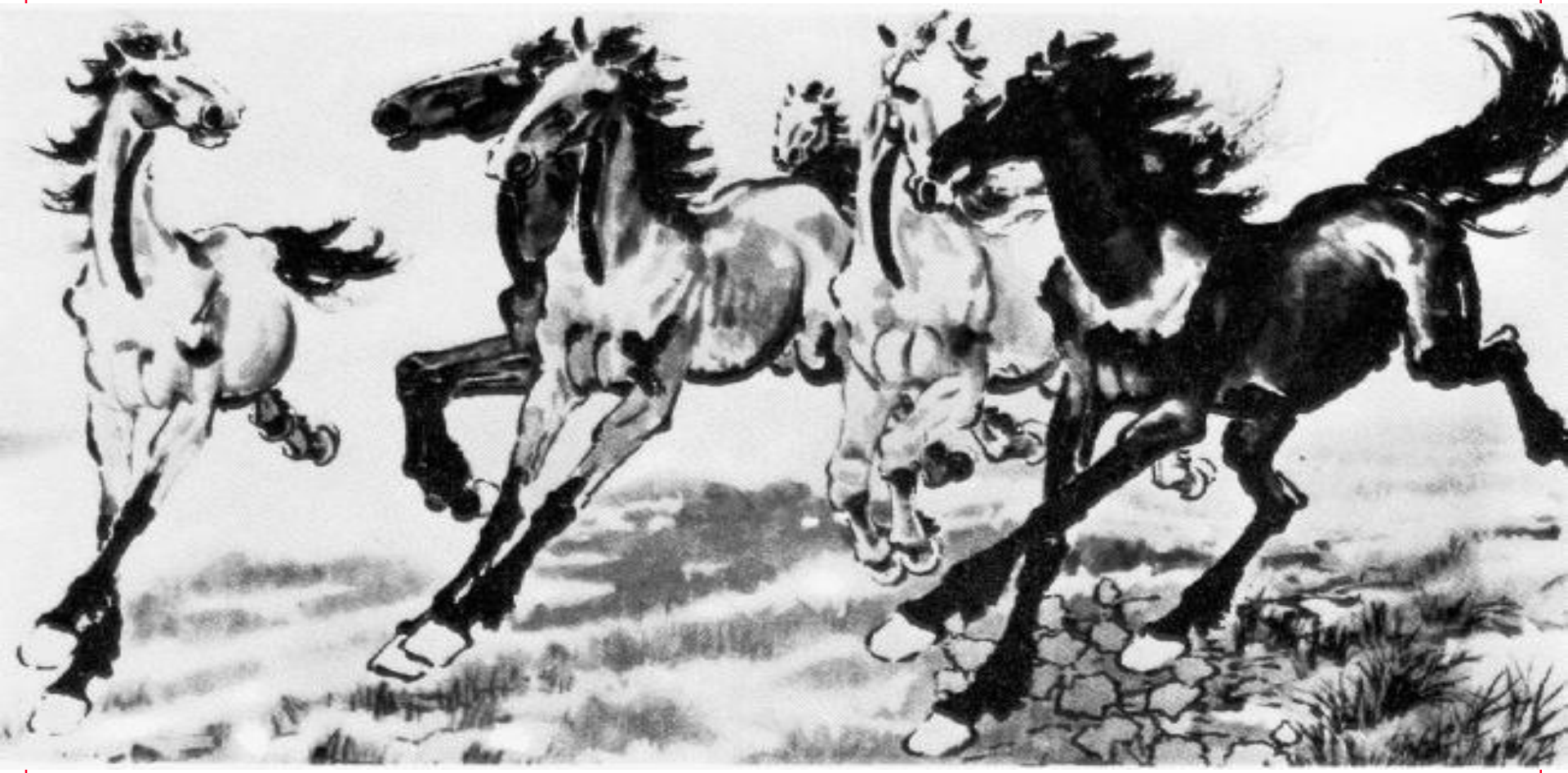


Magritte, 1957

Challenges 4: scale

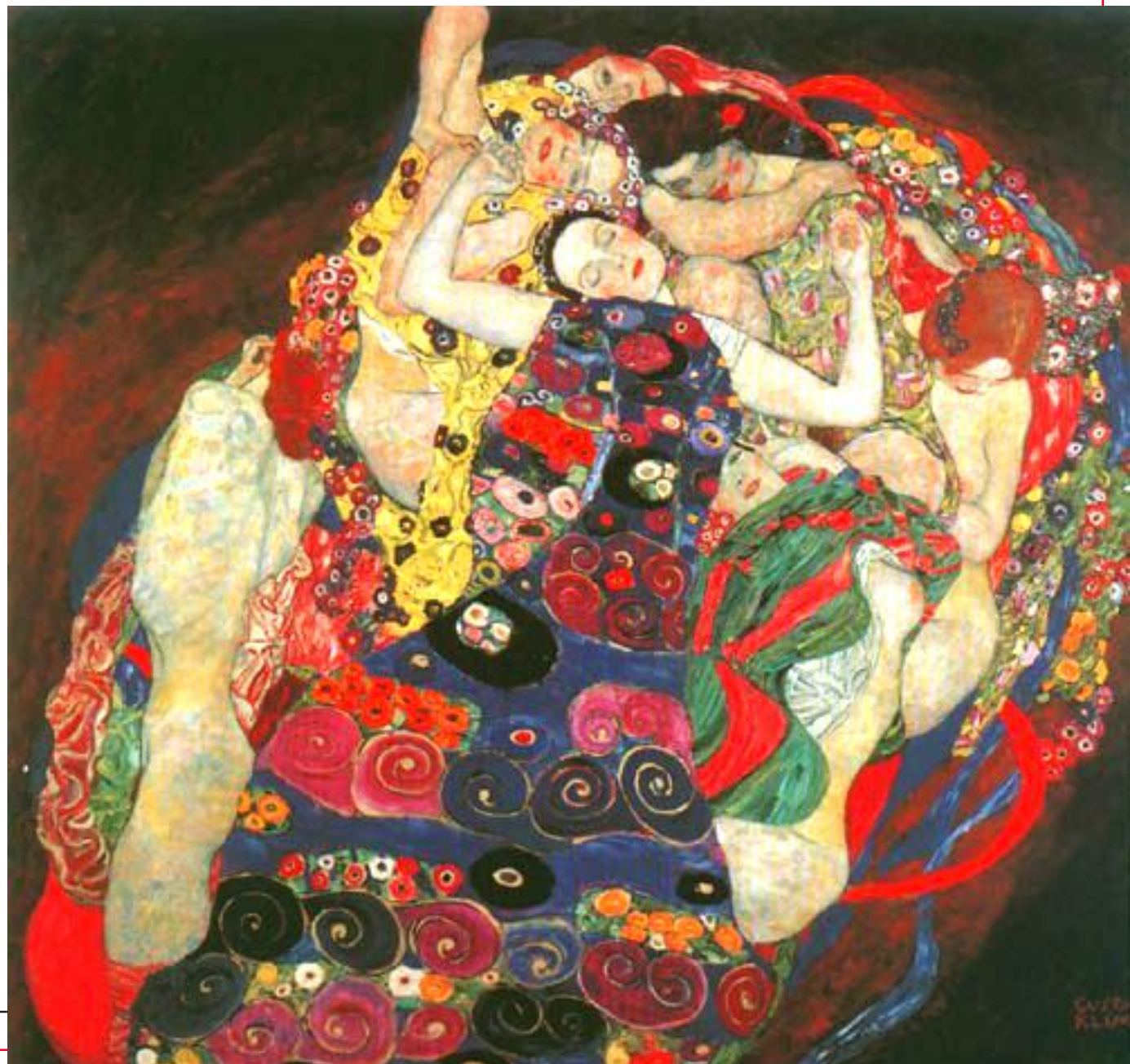


Challenges 5: deformation



Xu, Beihong 1943

Challenges 6: background clutter



Klimt, 1913

Challenges 6: background clutter



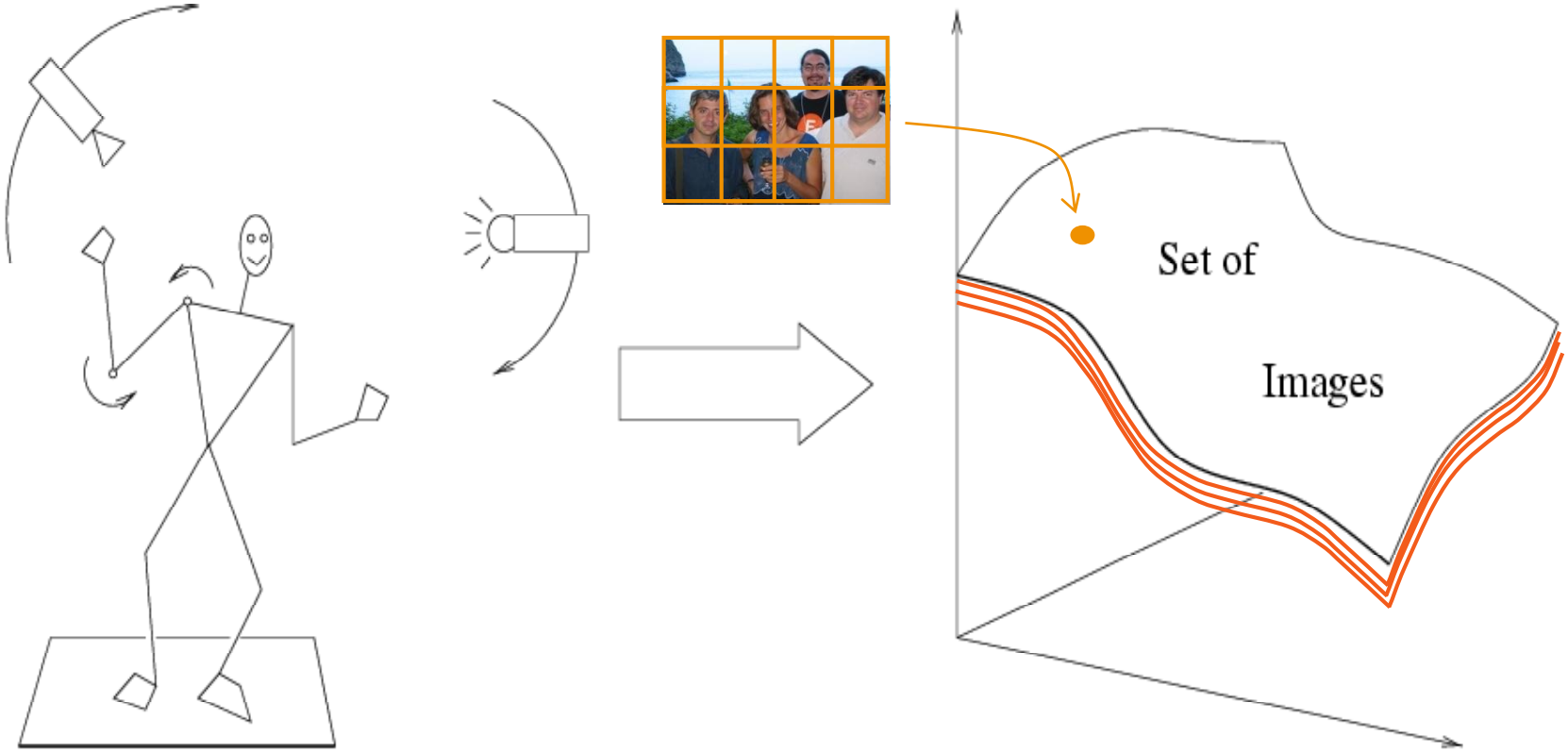
Brady, M. J., & Kersten, D. (2003). Bootstrapped learning of novel objects. J Vis, 3(6), 413-422

Challenges 6: background clutter



Brady, M. J., & Kersten, D. (2003). Bootstrapped learning of novel objects. J Vis, 3(6), 413-422

Modeling variability



Variability: Camera position
Illumination
Internal parameters

Within-class variations

Within-class variations











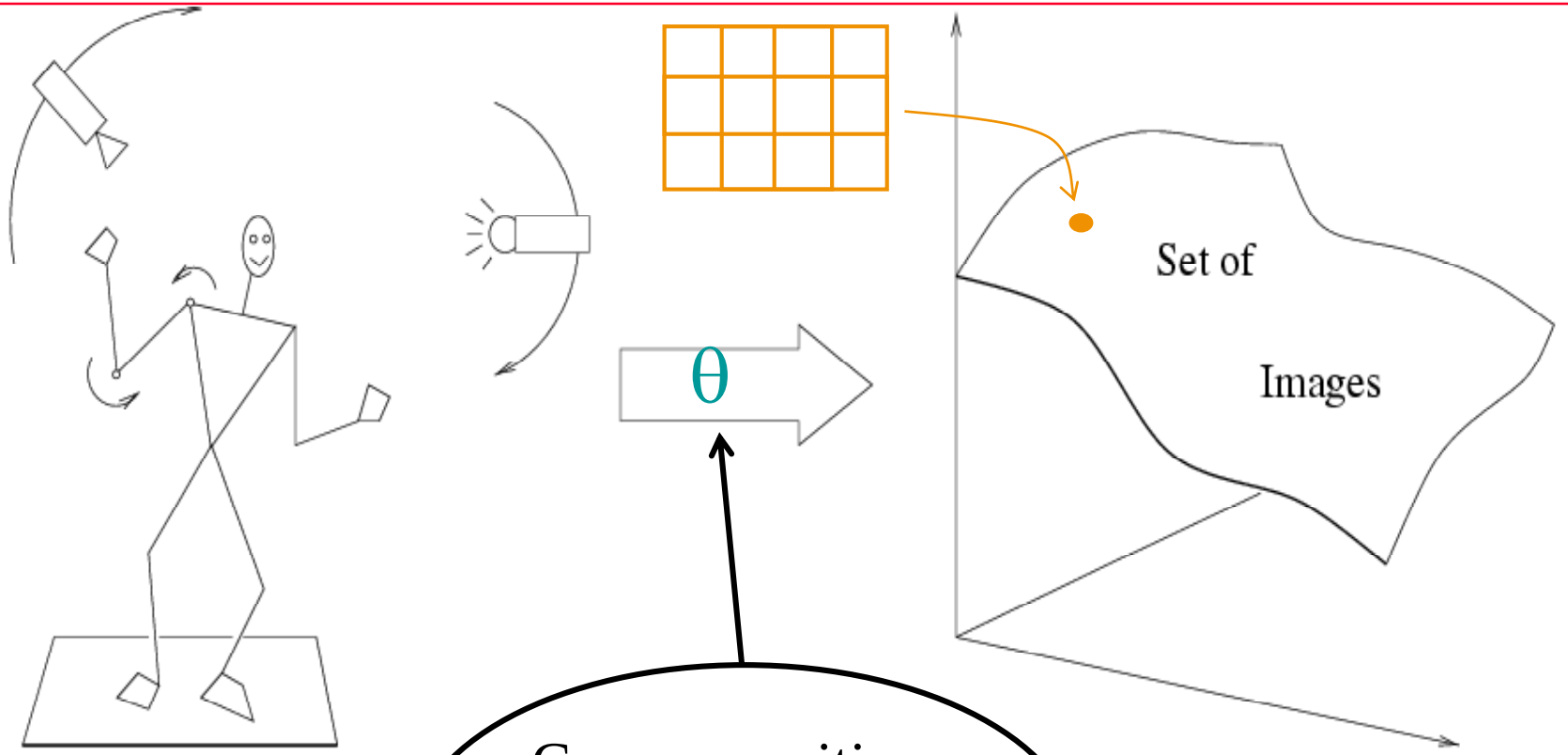


Within-class variations



Timeline of recognition

- 1965-late 1980s: alignment, geometric primitives



Variability:

Camera position
Illumination
Internal parameters

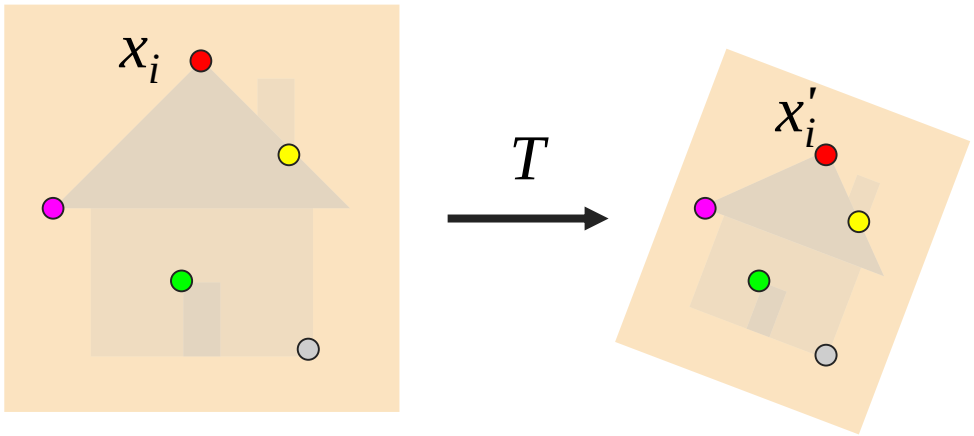
Alignment

Shape: assumed known

Roberts (1965); Lowe (1987); Faugeras & Hebert (1986); Grimson & Lozano-Perez (1986); Huttenlocher & Ullman (1987)

Recall: Alignment

- Alignment: fitting a model to a transformation between pairs of features (*matches*) in two images

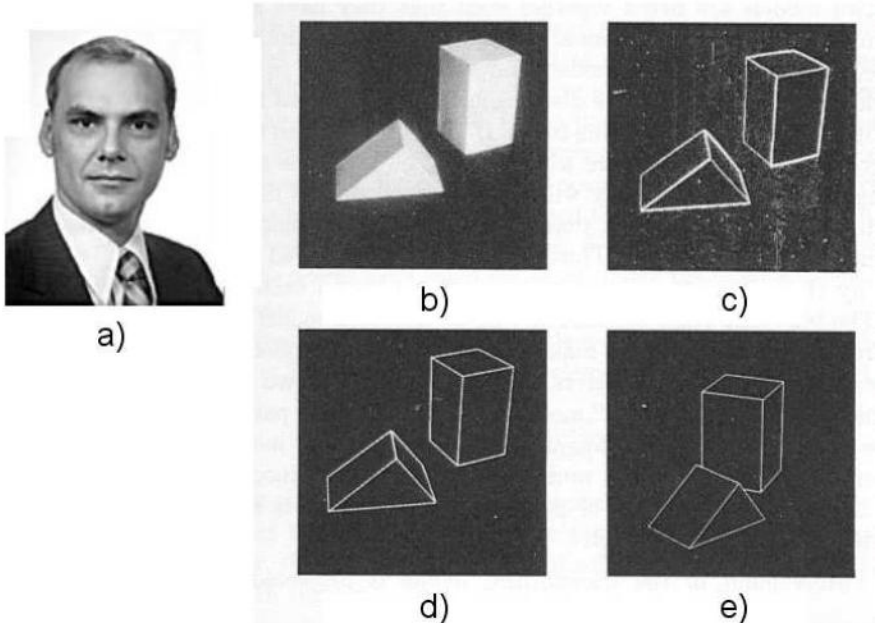


Find transformation T that minimizes

$$\sum_i \text{residual}(T(x_i), x_i')$$

提取特征点
建立特征点之间的对应关系
求解几何变换参数

Recognition as an alignment problem: Block world

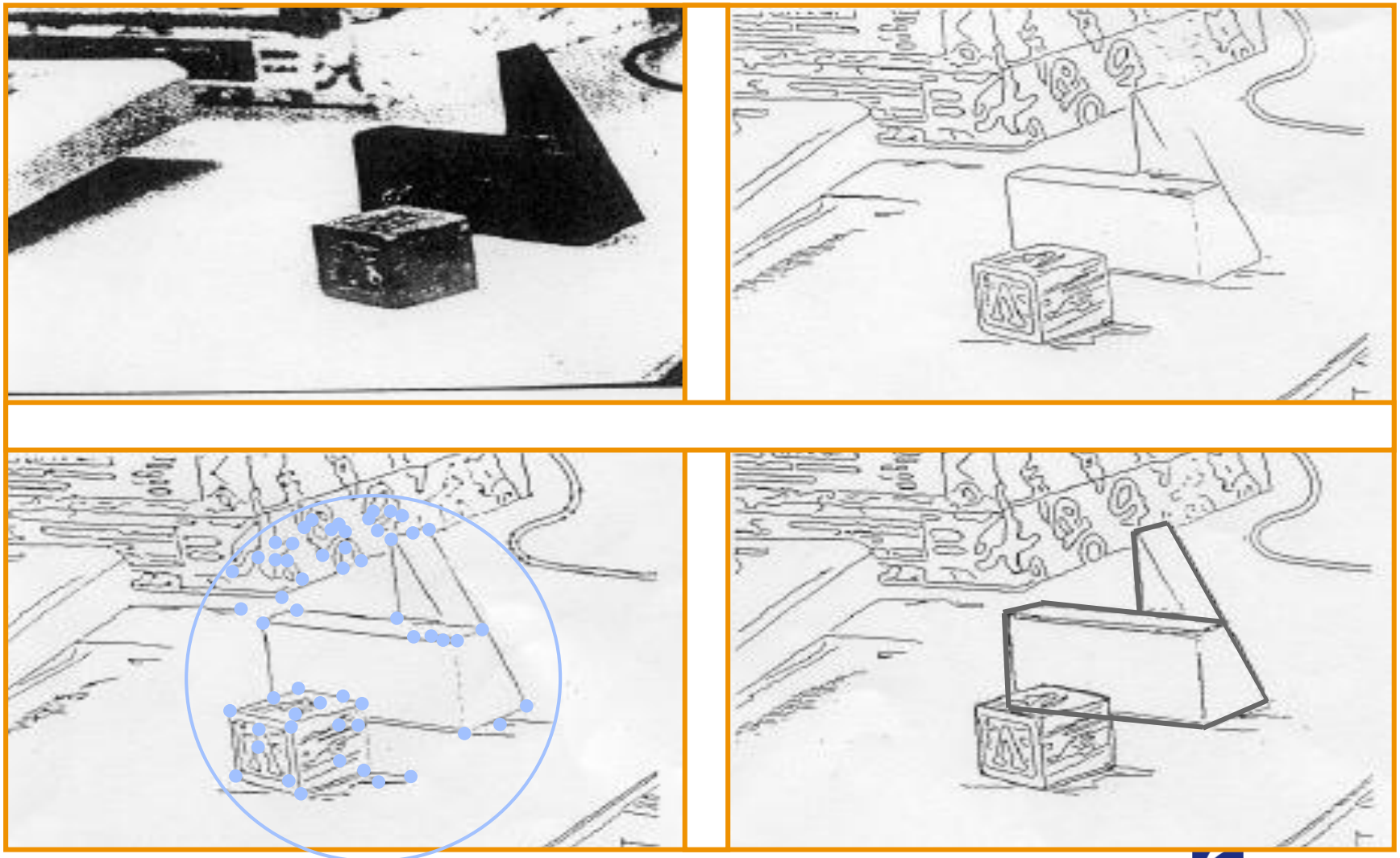


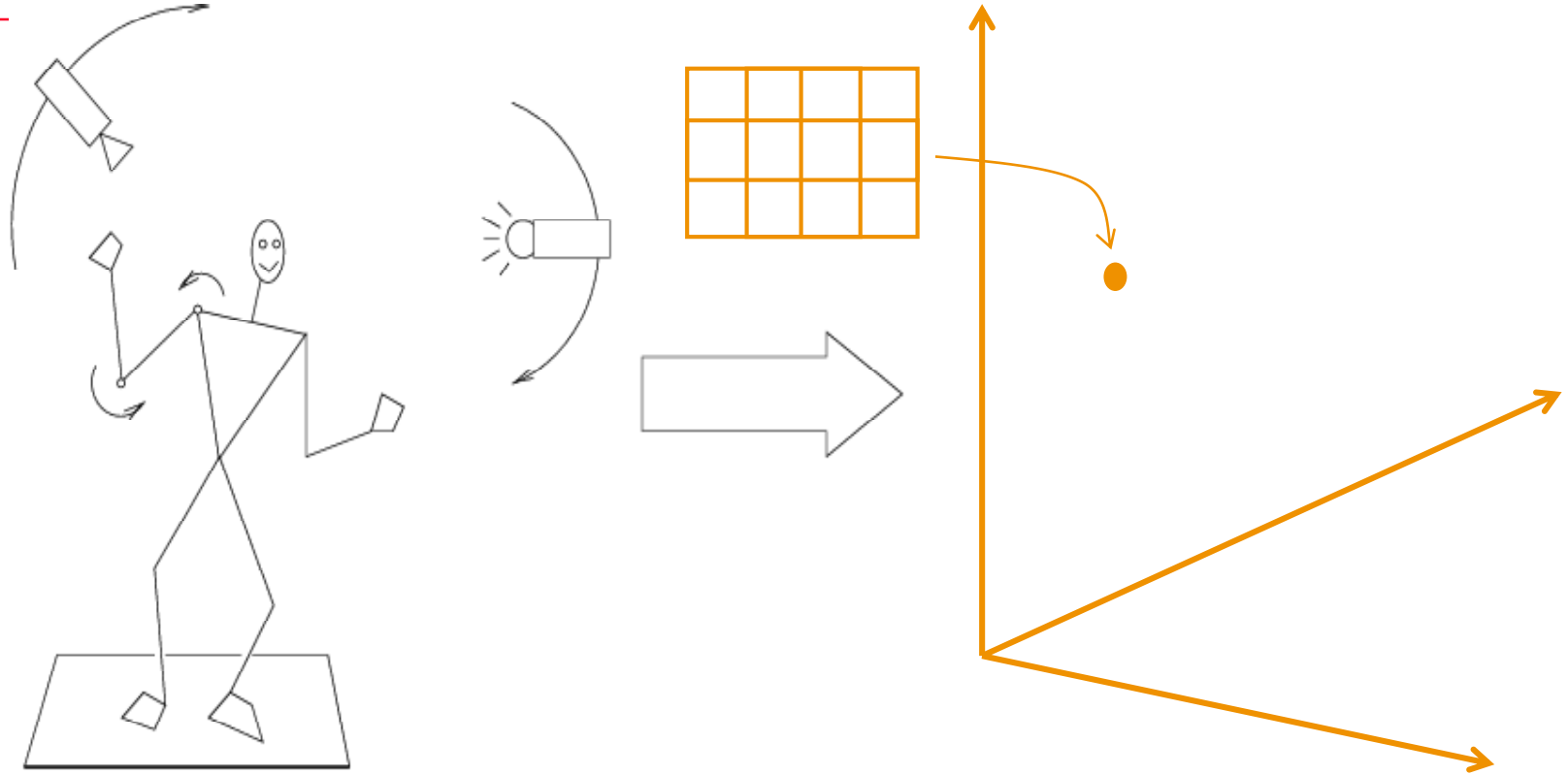
L. G. Roberts, *Machine Perception of Three Dimensional Solids*, Ph.D. thesis, MIT Department of Electrical Engineering, 1963.

Fig. 1. A system for recognizing 3-d polyhedral scenes. a) L.G. Roberts. b)A blocks world scene. c)Detected edges using a 2x2 gradient operator. d) A 3-d polyhedral description of the scene, formed automatically from the single image. e) The 3-d scene displayed with a viewpoint different from the original image to demonstrate its accuracy and completeness. (b) - e) are taken from [64] with permission MIT Press.)

Nice framework to develop fancy math, but too far from reality…

Alignment: Huttenlocher & Ullman (1987)





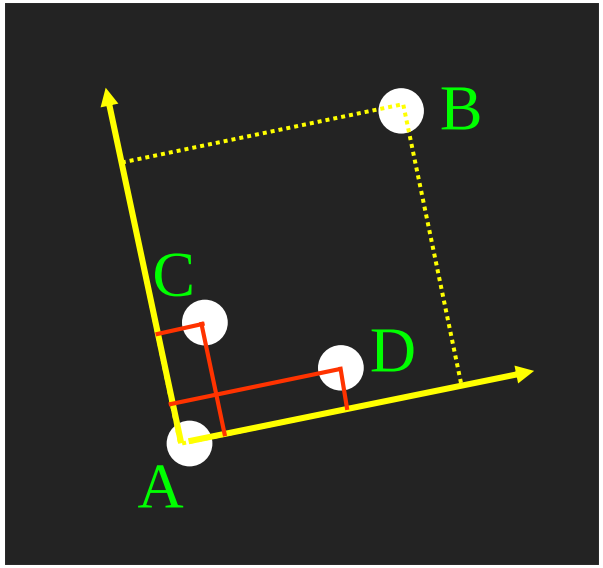
~~Variability~~

Invariance to:

Camera position
Illumination
Internal parameters

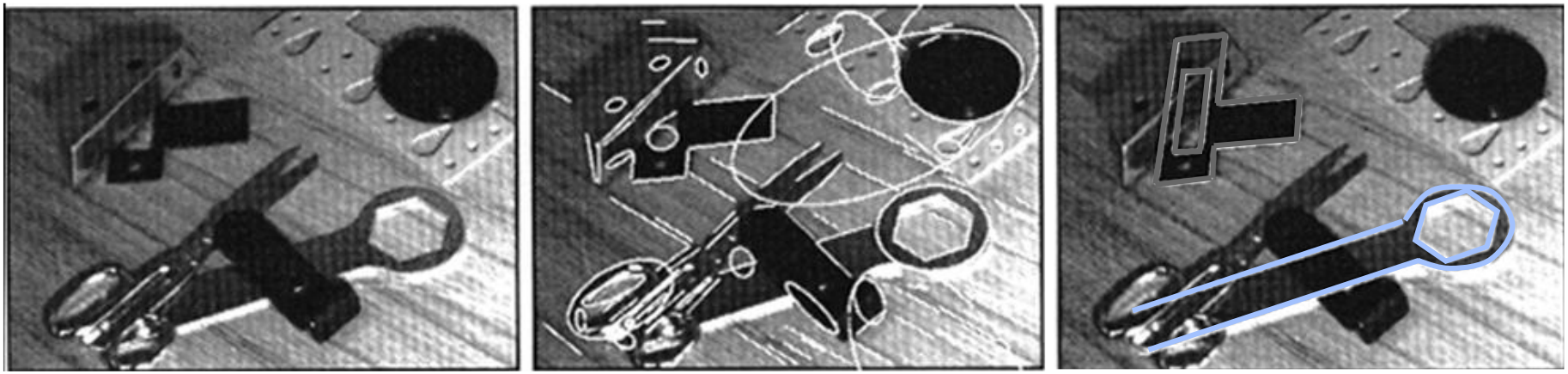
Duda & Hart (1972); Weiss (1987); Mundy et al. (1992-94);
Rothwell et al. (1992); Burns et al. (1993)

Example: invariant to similarity transformations computed from four points



交比 cross ratio

Projective invariants (Rothwell et al., 1992):



General 3D objects do not admit monocular viewpoint invariants (Burns et al., 1993)

Recognition by components



Irving Biederman
Recognition-by-Components: A Theory of Human Image Understanding.
Psychological Review, 1987.

比德曼：在心理物理学和计算机视觉方面做出了很多有意义的成果
1987年，提出了基于“成分”的目标识别，目前很多计算机视觉模型都是使用这种思路

Recognition by components

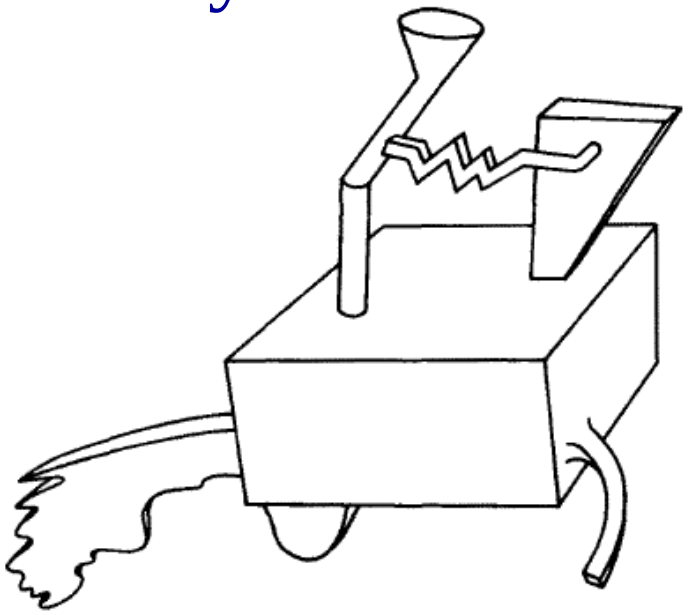
The fundamental assumption of the proposed theory, recognition-by-components (RBC), is that a modest set of generalized-cone components, called geons (*N* = 36), can be derived from contrasts of five readily detectable properties of edges in a two-dimensional image: curvature, collinearity, symmetry, parallelism, and cotermination.

The "contribution lies in its proposal for a particular vocabulary of components derived from perceptual mechanisms and its account of how an arrangement of these components can access a representation of an object in memory."

基于成分的目标识别方法中，基本成分包括二维图像中边缘的曲率、共线性、对称性、相似性、同时终止等五个属性

该方法的主要贡献：可以构建字典，通过字典的排列近似目标

A do-it-yourself example



（1）首先这个物体不是我们已知物体

（2）我们可以用大家公认的方法把目标分为不同的部分

（3）我们可以知道跟我们熟悉东西的相似性：热狗车

1) We know that this object is nothing we know
2) We can split this objects into parts that everybody will agree
3) We can see how it resembles something familiar: “a hot dog cart”

“The naive realism that emerges in descriptions of nonsense objects may be reflecting the workings of a representational system by which objects are identified.”

Hypothesis

- Hypothesis: there is a small number of geometric components that constitute the primitive elements of the object recognition system (like letters to form words).

  假设目标识别中仅有少数的几何基元（类似于单词仅有少数字母组成）

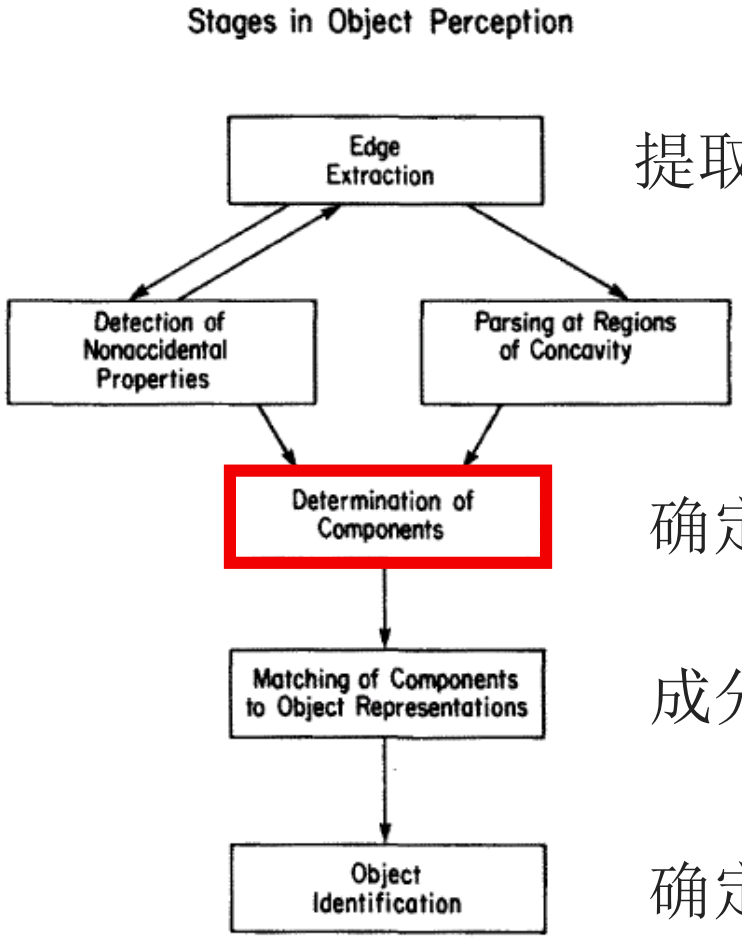
- “The particular properties of edges that are postulated to be relevant to the generation of the volumetric primitives have the desirable properties that they are invariant over changes in orientation and can be determined from just a few points on each edge.”

  希望提取对方向不敏感的基元，可以由边缘上的少数点确定

- Limitation: “The modeling has been limited to concrete entities with specified boundaries.” (count nouns) – this limitation is shared by many modern object detection algorithms.

  受限于具体实体的特定边缘，这也是目前一些目标检测算法的局限性

Stages of processing



检测重要属性

提取边缘

解析凹区域

确定成分

成分与目标的匹配

确定目标

Figure 2. Presumed processing stages in object recognition.

"Parsing is performed, primarily at concave regions, simultaneously with a detection of nonaccidental properties."

Examples:

- Colinearity
- Smoothness
- Symmetry
- Parallelism
- Cotermination

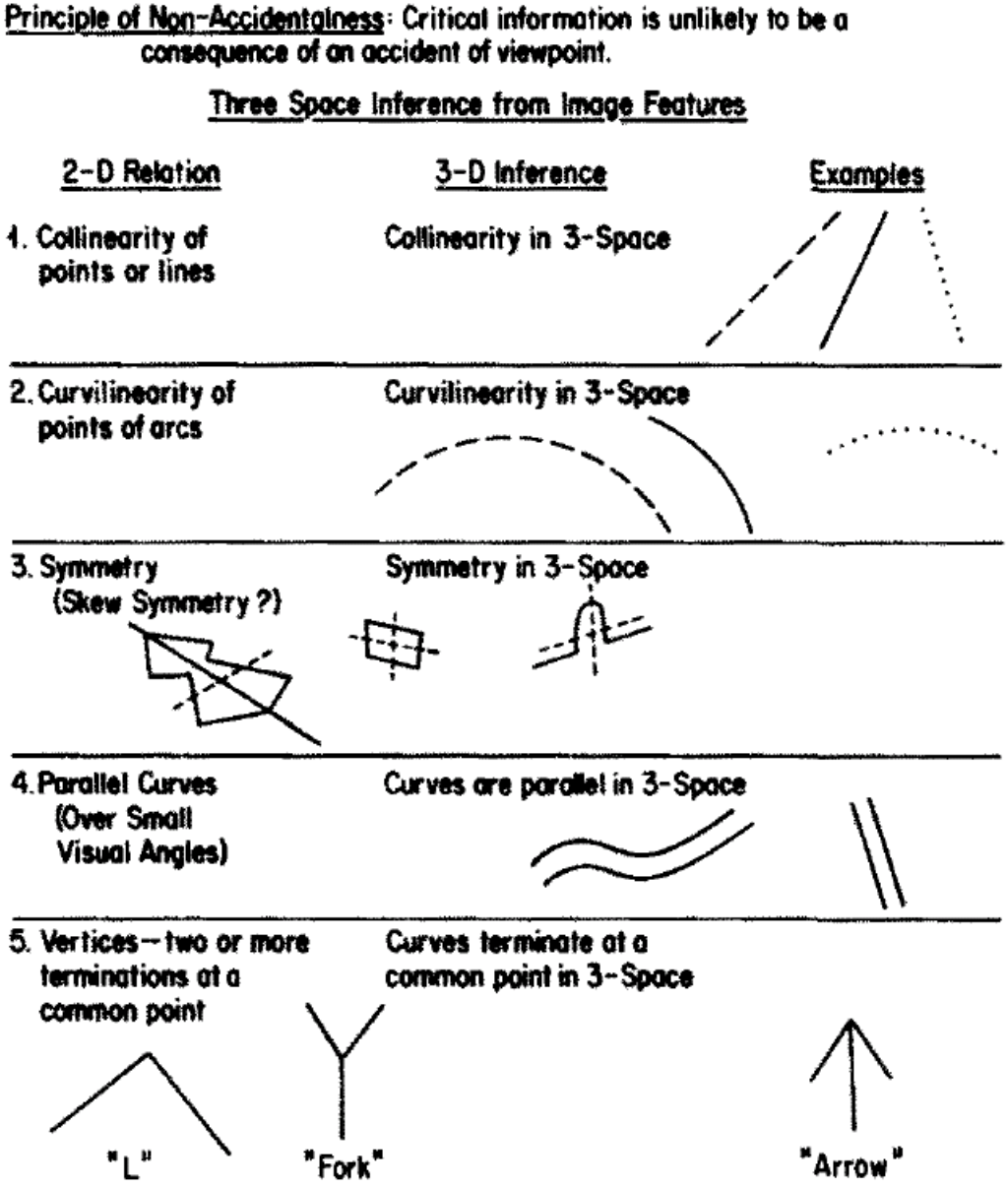
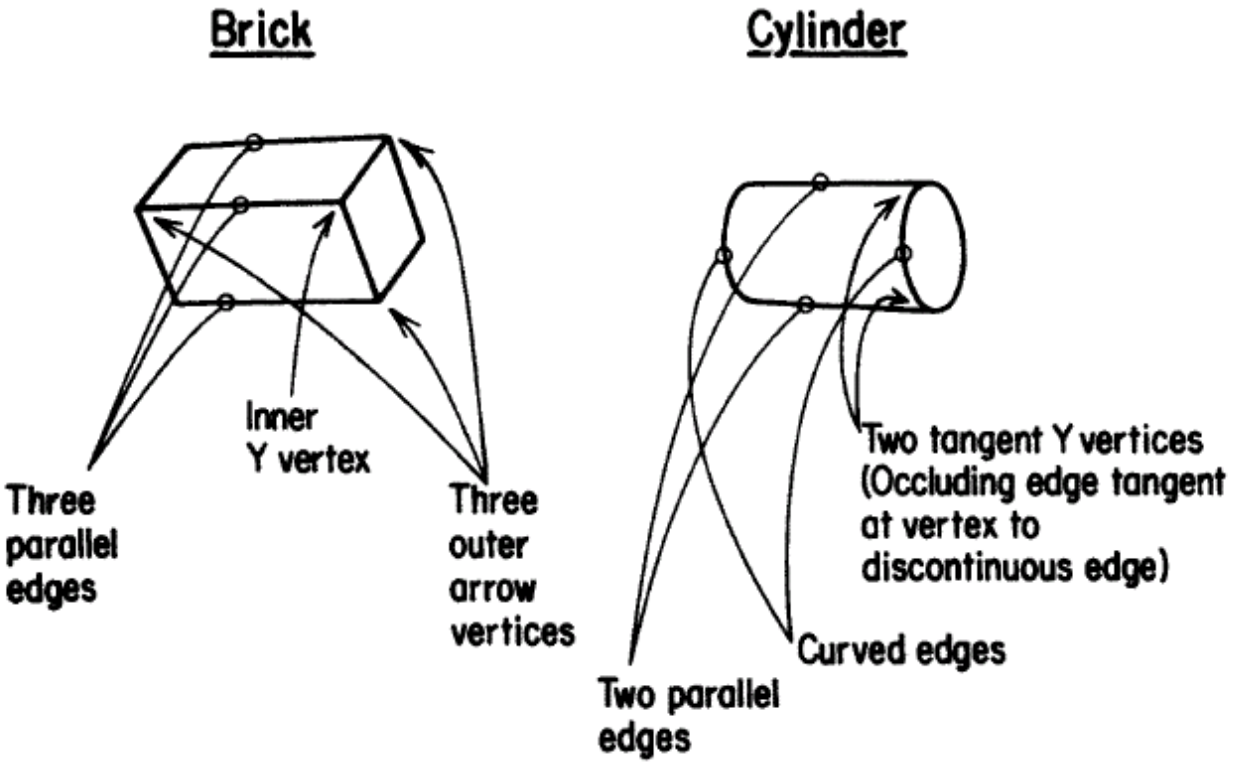


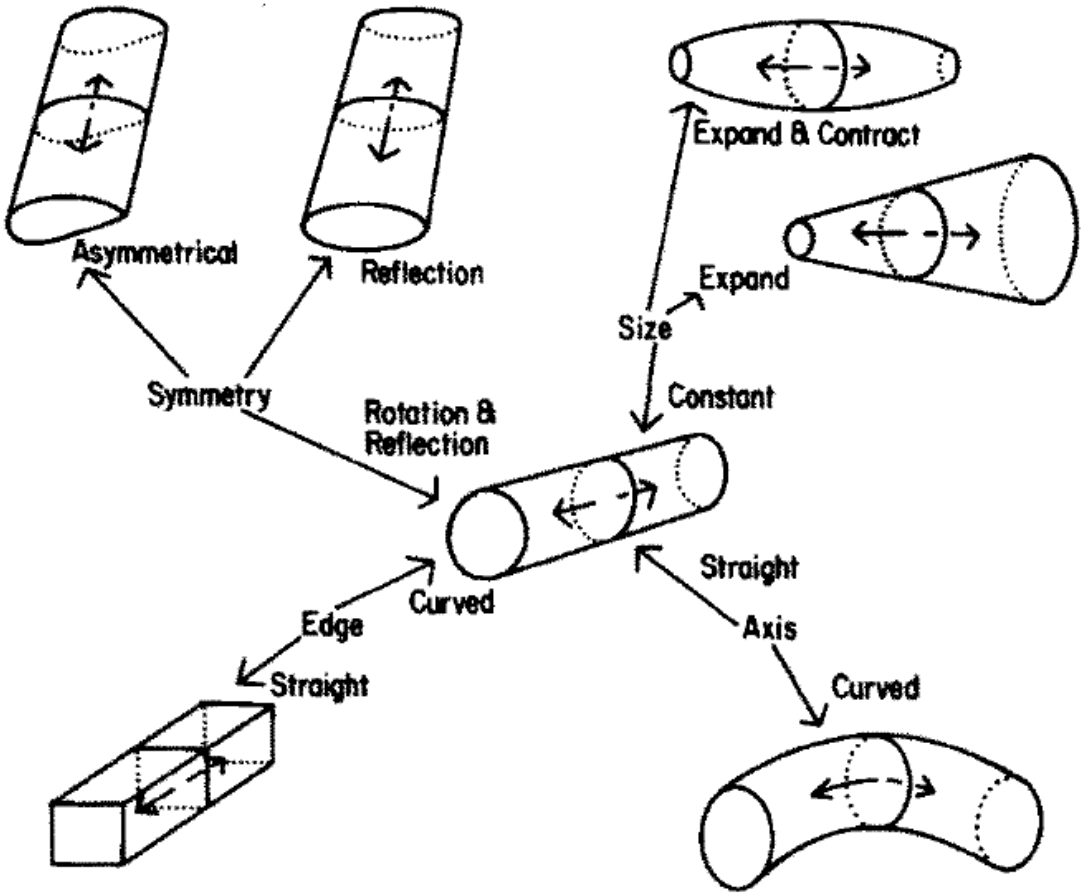
Figure 4. Five nonaccidental relations. (From Figure 5.2, *Perceptual organization and visual recognition* [p. 77] by David Lowe. Unpublished doctorial dissertation, Stanford University. Adapted by permission.)

Some Nonaccidental Differences Between a Brick and a Cylinder



The high speed and accuracy of determining a given nonaccidental relation {e.g., whether some pattern is symmetrical) should be contrasted with performance in making absolute quantitative judgments of variations in a single physical attribute,
such as length of a segment or degree of tilt or curvature.
Object recognition is performed by humans in around 100ms.

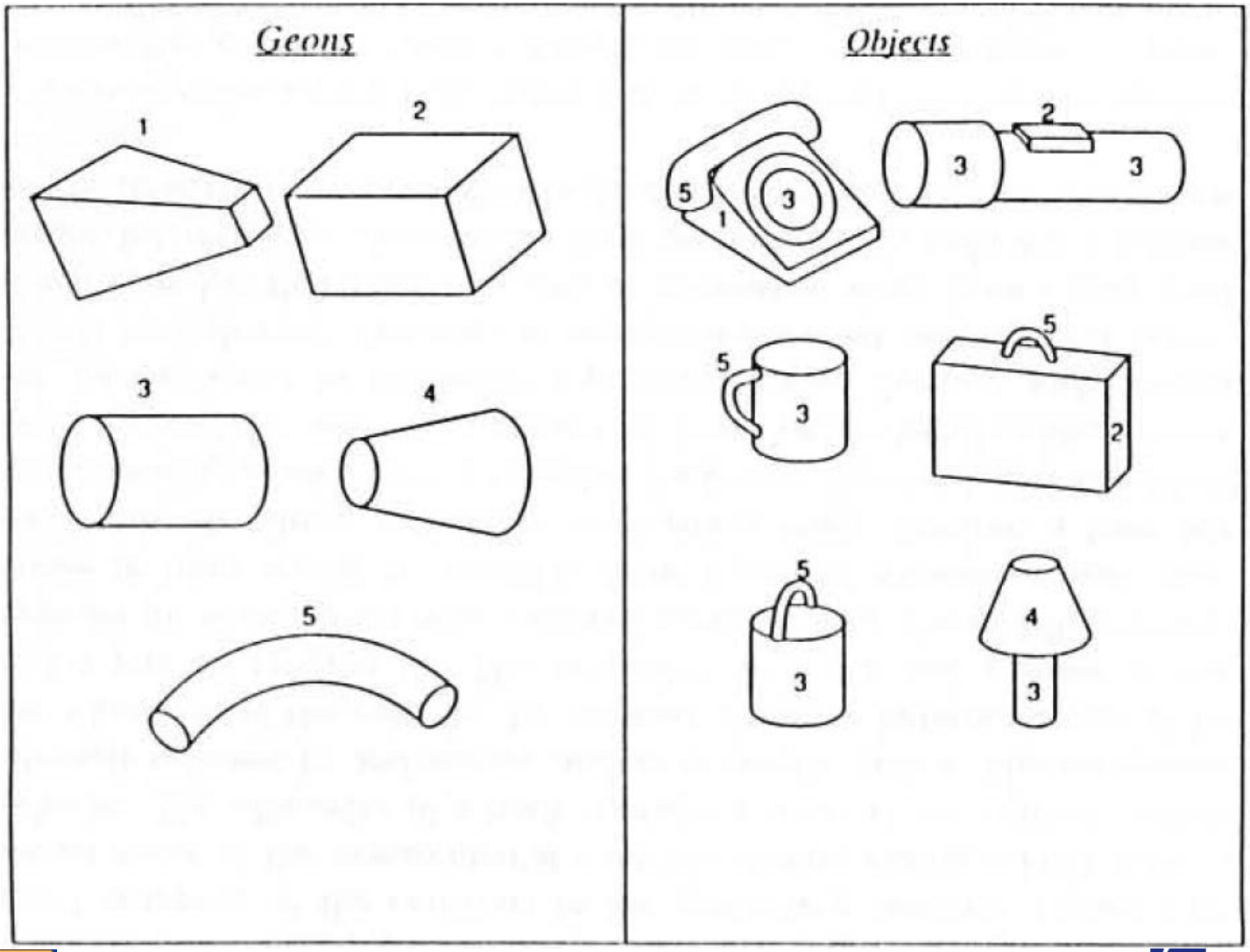
From generalized cylinders to GEONS



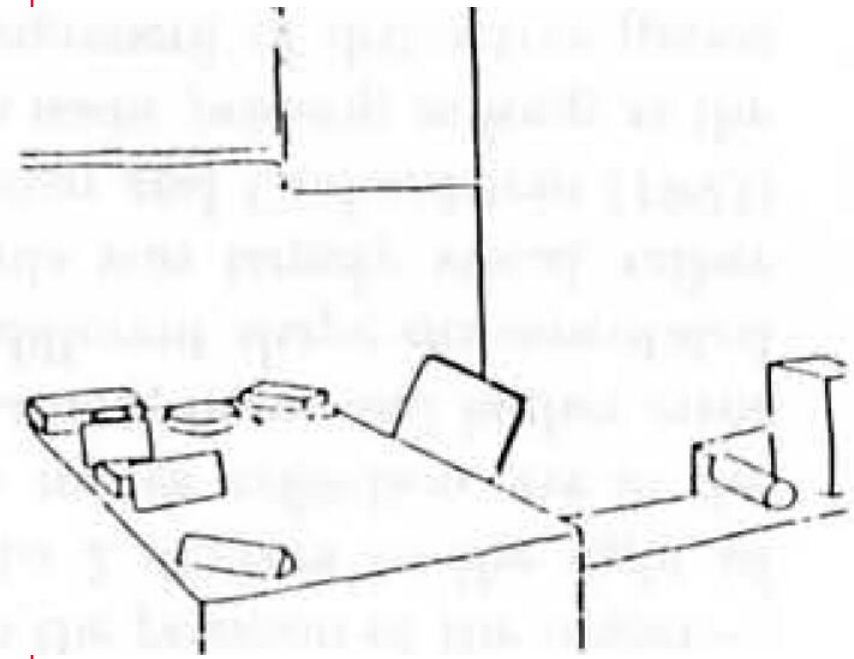
"From variation over only two or three levels in the nonaccidental relations of four attributes of generalized cylinders, a set of 36 GEONS can be generated."

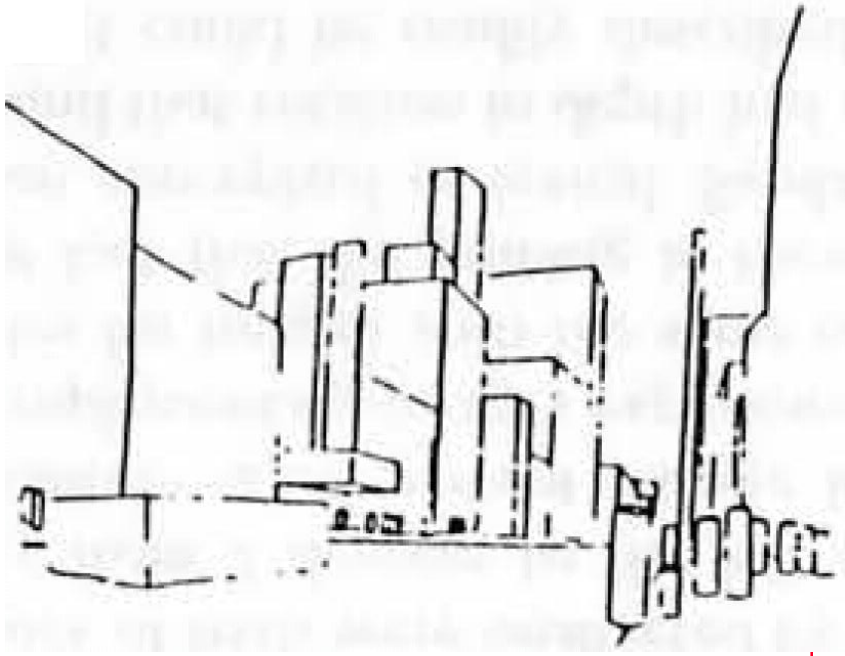
Geons represent a restricted form of generalized cylinders.

Objects and their geons



Scenes and geons





Mezzanotte & Biederman

The importance of spatial arrangement

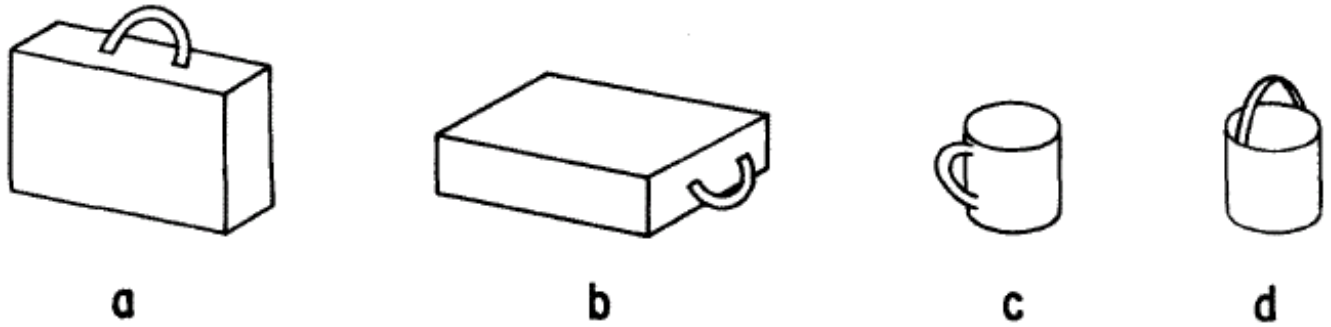
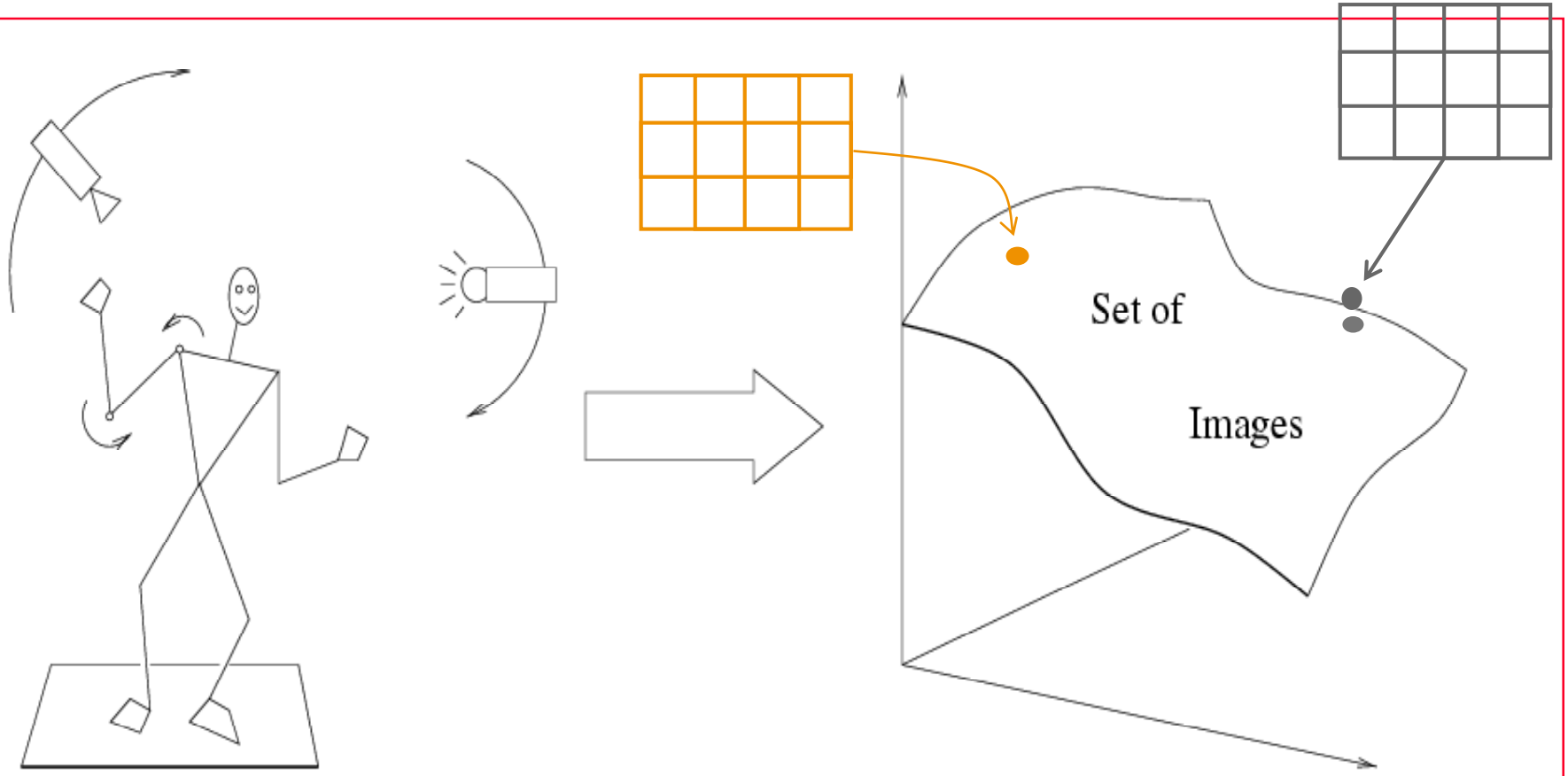


Figure 3. Different arrangements of the same components can produce different objects.

Timeline of recognition

- 1965-late 1980s: alignment, geometric primitives
- Early 1990s: invariants, appearance-based methods



Empirical models of image variability

Appearance-based techniques

Turk & Pentland (1991); Murase & Nayar (1995); etc.

Eigenfaces (Turk & Pentland, 1991)





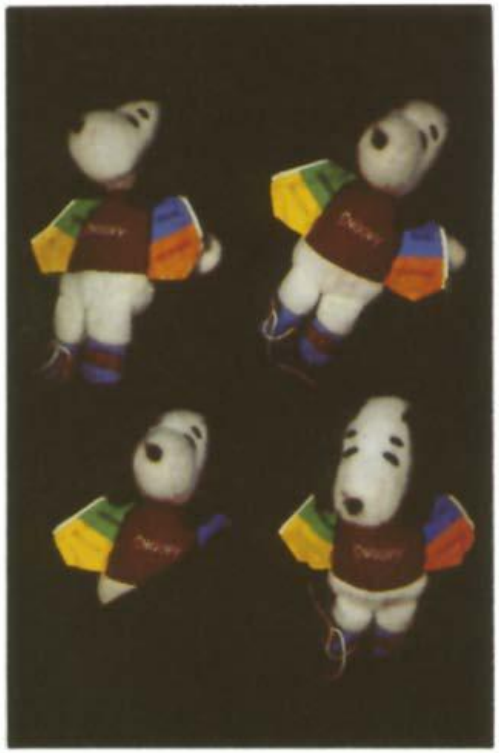


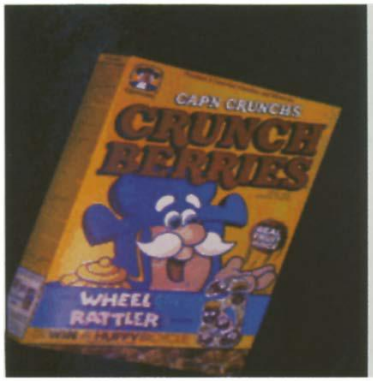


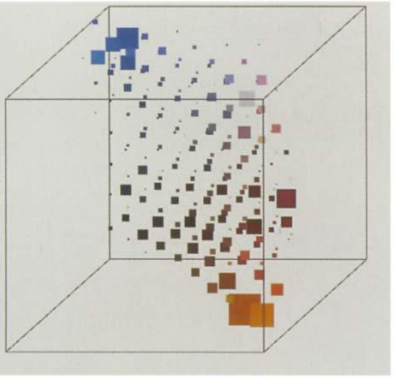


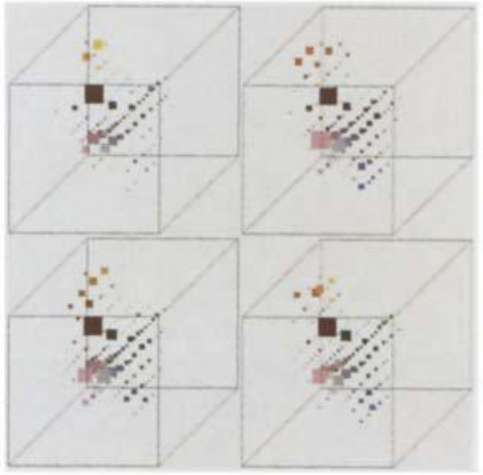
Experimental	Correct/Unknown Recognition Percentage		
Condition	Lighting	Orientation	Scale
Forced classification	96/0	85/0	64/0
Forced 100% accuracy	100/19	100/39	100/60
Forced 20% unknown rate	100/20	94/20	74/20

Color Histograms



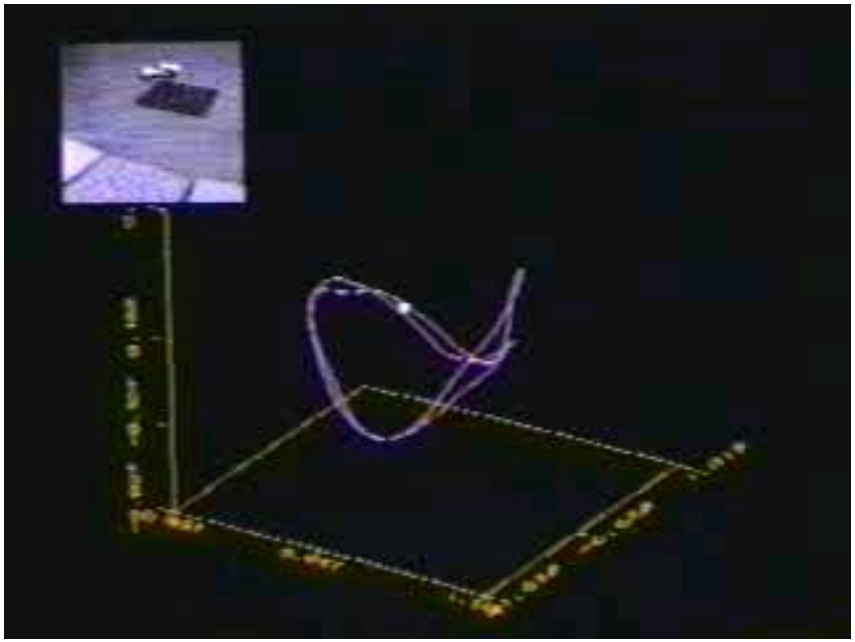






Swain and Ballard, Color Indexing, IJCV 1991.

Appearance manifolds





H. Murase and S. Nayar, Visual learning and recognition of 3-d objects from appearance, IJCV 1995

Limitations of global appearance models

- Can work on relatively simple patterns



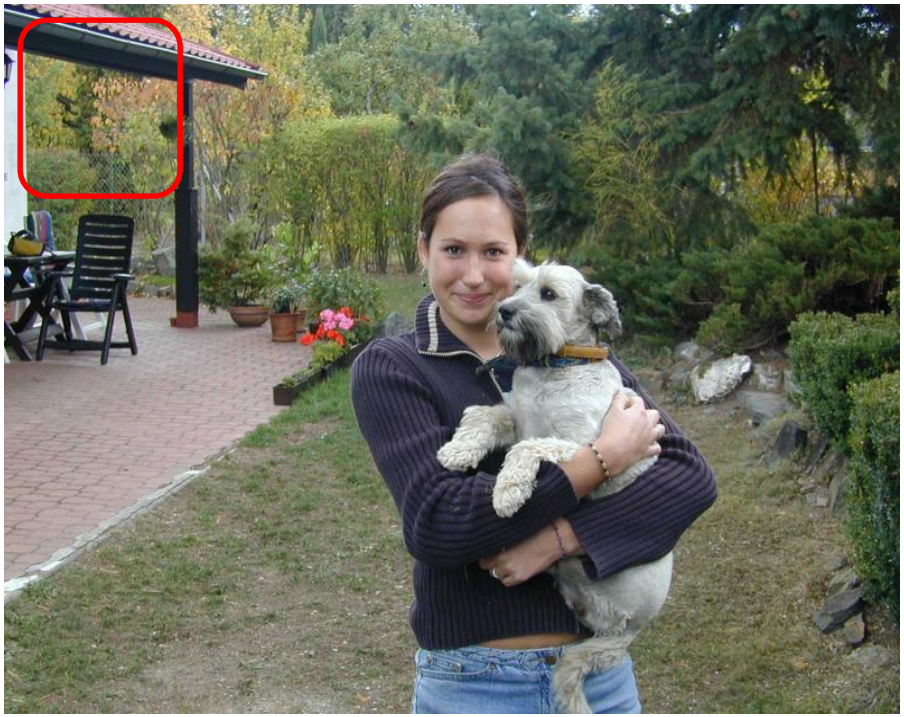
Timeline of recognition

- 1965-late 1980s: alignment, geometric primitives
- Early 1990s: invariants, appearance-based methods
- Mid-late 1990s: sliding window approaches

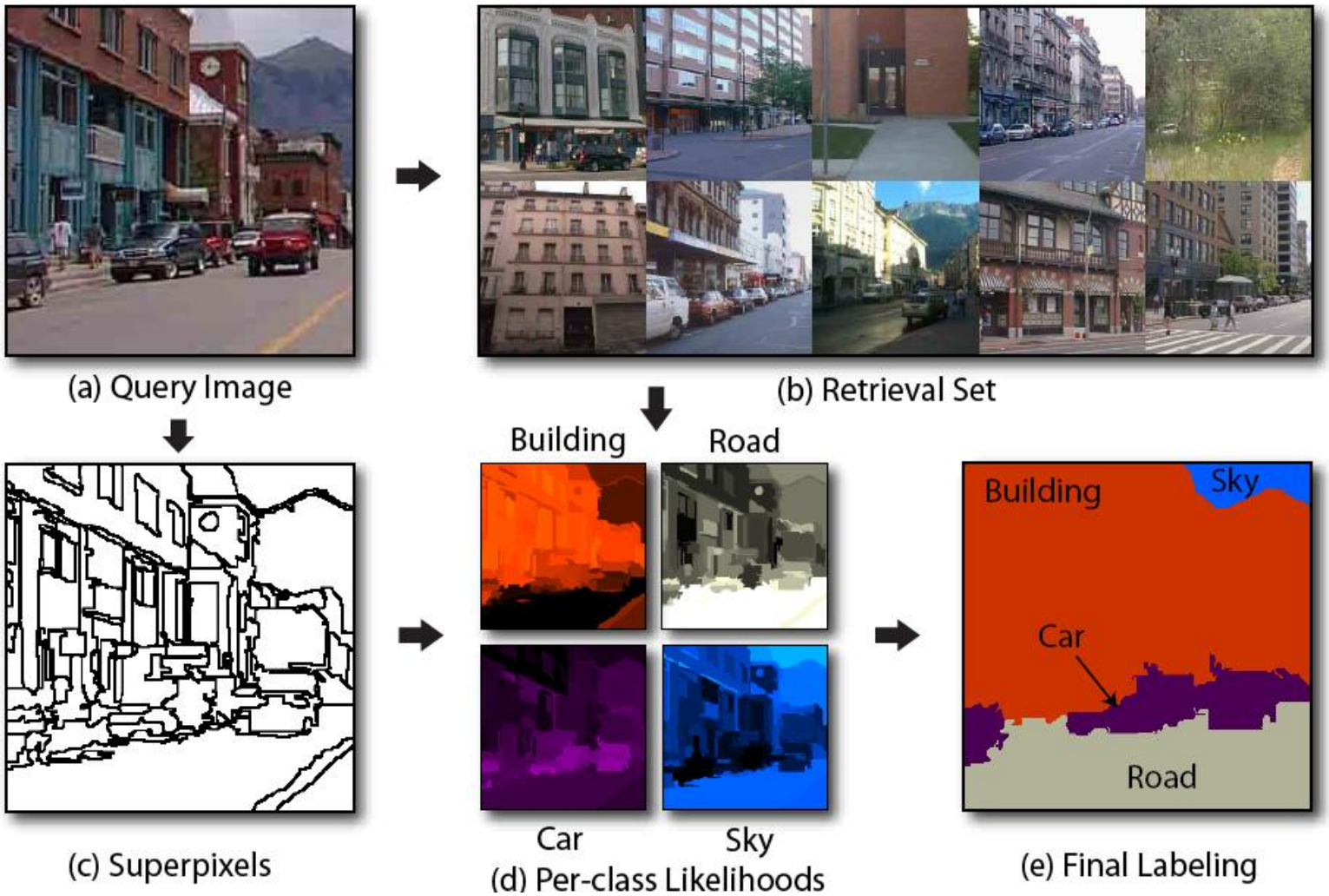
Sliding window approaches

Classify each window separately

Scale / orientation range to search over



Scene-level context for image parsing

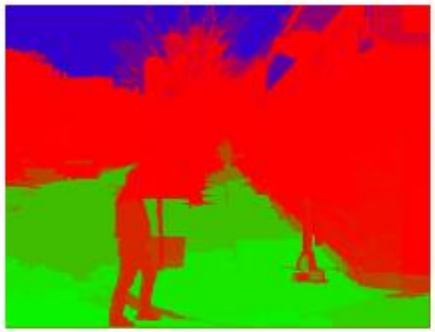


J. Tighe and S. Lazebnik, SuperParsing Scalable Nonparametric Image Parsing with Superpixels, ECCV 2010

Geometric context



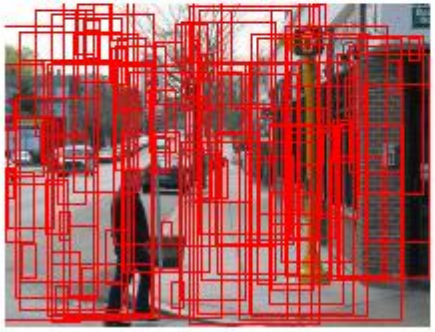
(a) Input image



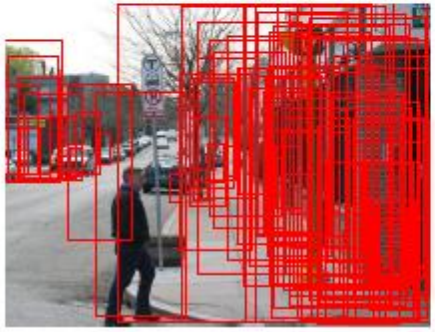
(c) Surface estimate



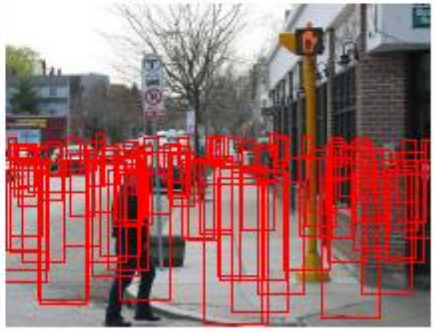
(e) P(viewpoint | objects)



(b) P(person) = uniform



(d) P(person | geometry)



(f) P(person | viewpoint)



(g) P(person|viewpoint,geometry)

D. Hoiem, A. Efros, and M. Herbert. Putting Objects in Perspective. CVPR 2006.

Timeline of recognition

- 1965-late 1980s: alignment, geometric primitives
- Early 1990s: invariants, appearance-based methods
- Mid-late 1990s: sliding window approaches
- Late 1990s: feature-based methods

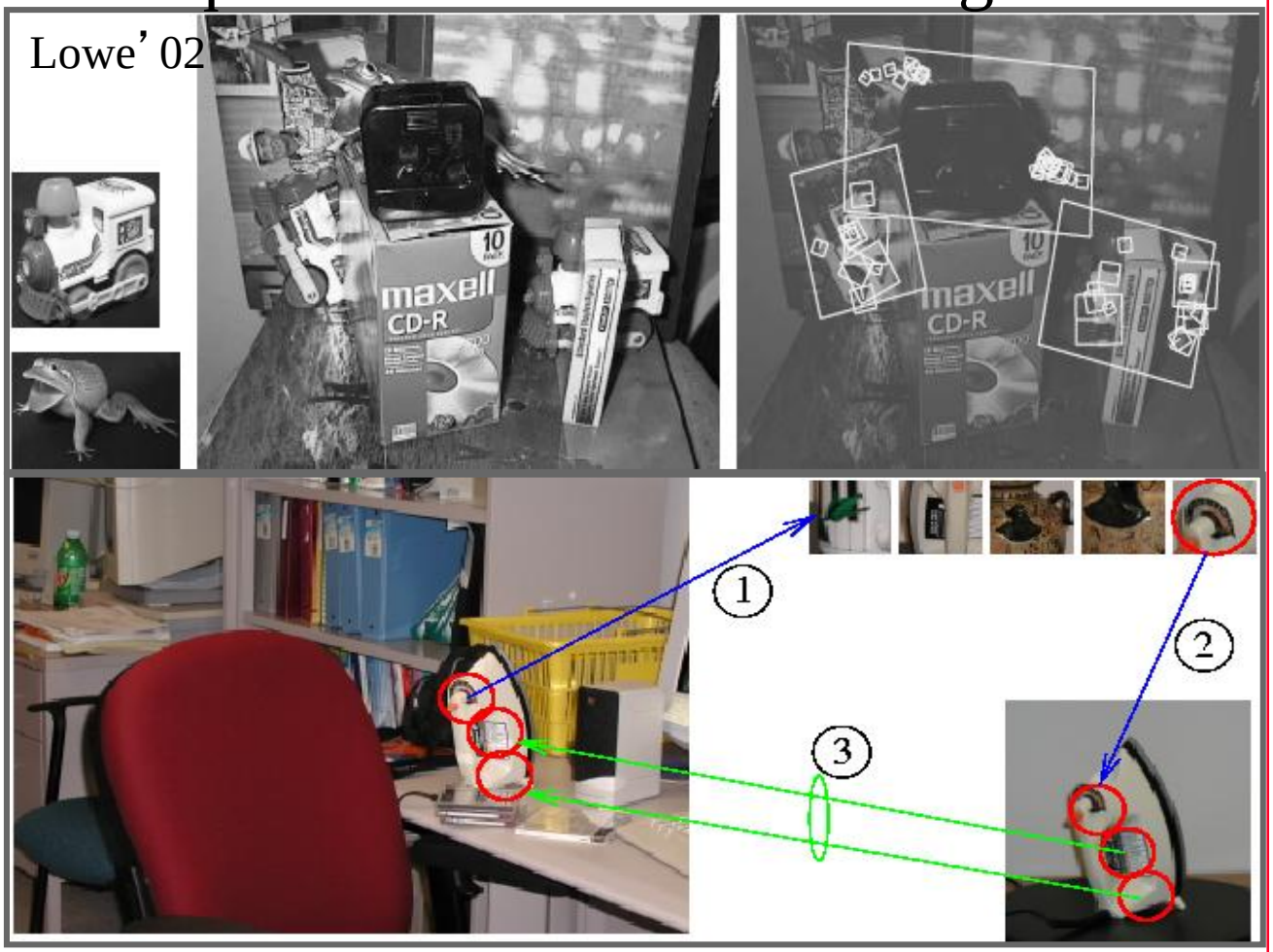
Local features

Combining *local* appearance, spatial constraints, invariants, and classification techniques from machine learning.





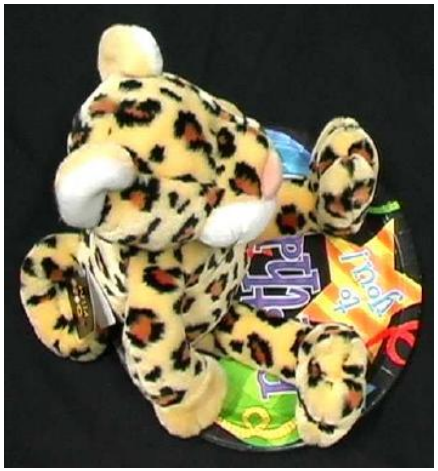
Schmid & Mohr' 97



Mahamud & Hebert' 03

Specific Object Recognition









Timeline of recognition

- 1965-late 1980s: alignment, geometric primitives
- Early 1990s: invariants, appearance-based methods
- Mid-late 1990s: sliding window approaches
- Late 1990s: feature-based methods
- Early 2000s – present : parts-and-shape models

Parts and Structure approaches

With a different perspective, these models focused more on the geometry than on defining the constituent elements:

- Fischler & Elschlager 1973
- Yuille ‘91
- Brunelli & Poggio ‘93
- Lades, v.d. Malsburg et al. ‘93
- Cootes, Lanitis, Taylor et al. ‘95
- Amit & Geman ‘95, ‘99
- Perona et al. ‘95, ‘96, ’98, ’00, ’03, ‘04
- Felzenszwalb & Huttenlocher ’00, ’04
- Crandall & Huttenlocher ’05, ’06
- Leibe & Schiele ’03, ’04
- Many papers since 2000

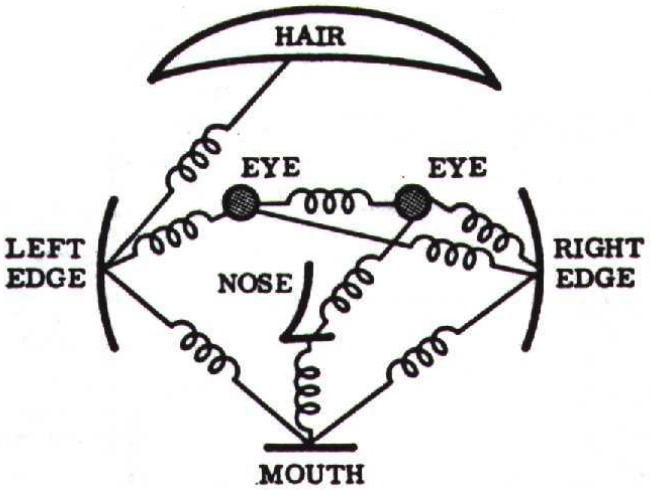


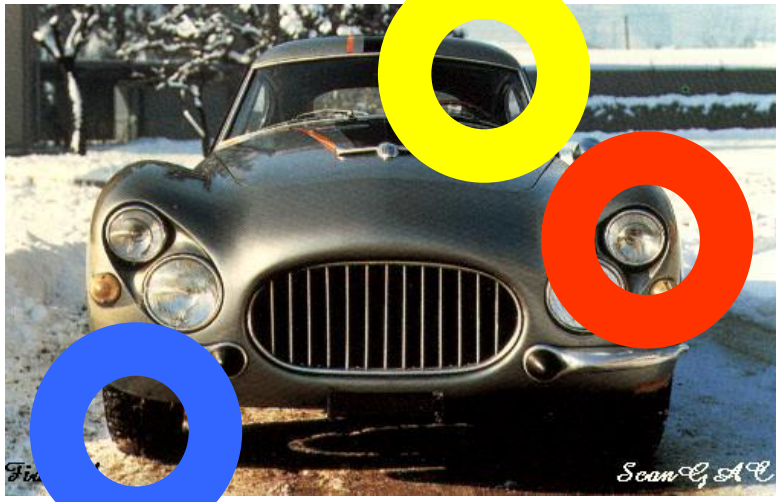
Figure from [Fischler & Elschlager 73]

Representing categories: Parts and Structure





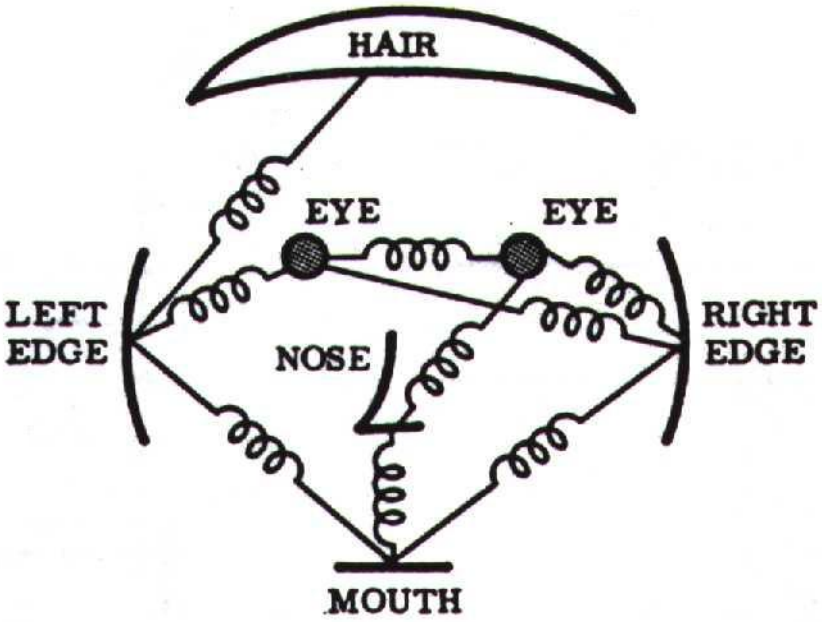




Weber, Welling & Perona (2000), Fergus, Perona & Zisserman (2003)

Representation

- Object as set of parts
  - Generative representation
- Model:
  - Relative locations between parts
  - Appearance of part
- Issues:
  - How to model location
  - How to represent appearance
  - Sparse or dense (pixels or regions)
  - How to handle occlusion/clutter



We will discuss these models more in depth next week

Timeline of recognition

- 1965-late 1980s: alignment, geometric primitives
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- Mid-late 1990s: sliding window approaches
- Late 1990s: feature-based methods
- Early 2000s – present : parts-and-shape models
- 2003 – present: bags of features

Bag-of-features models

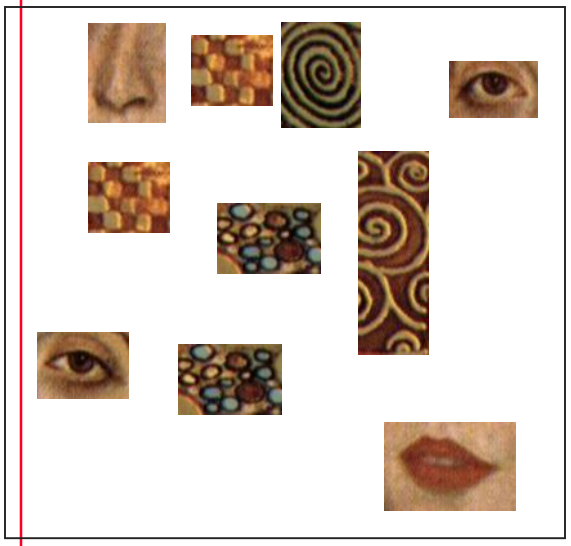
Object → **Bag of ‘words’**





Objects as texture

- All of these are treated as being the same







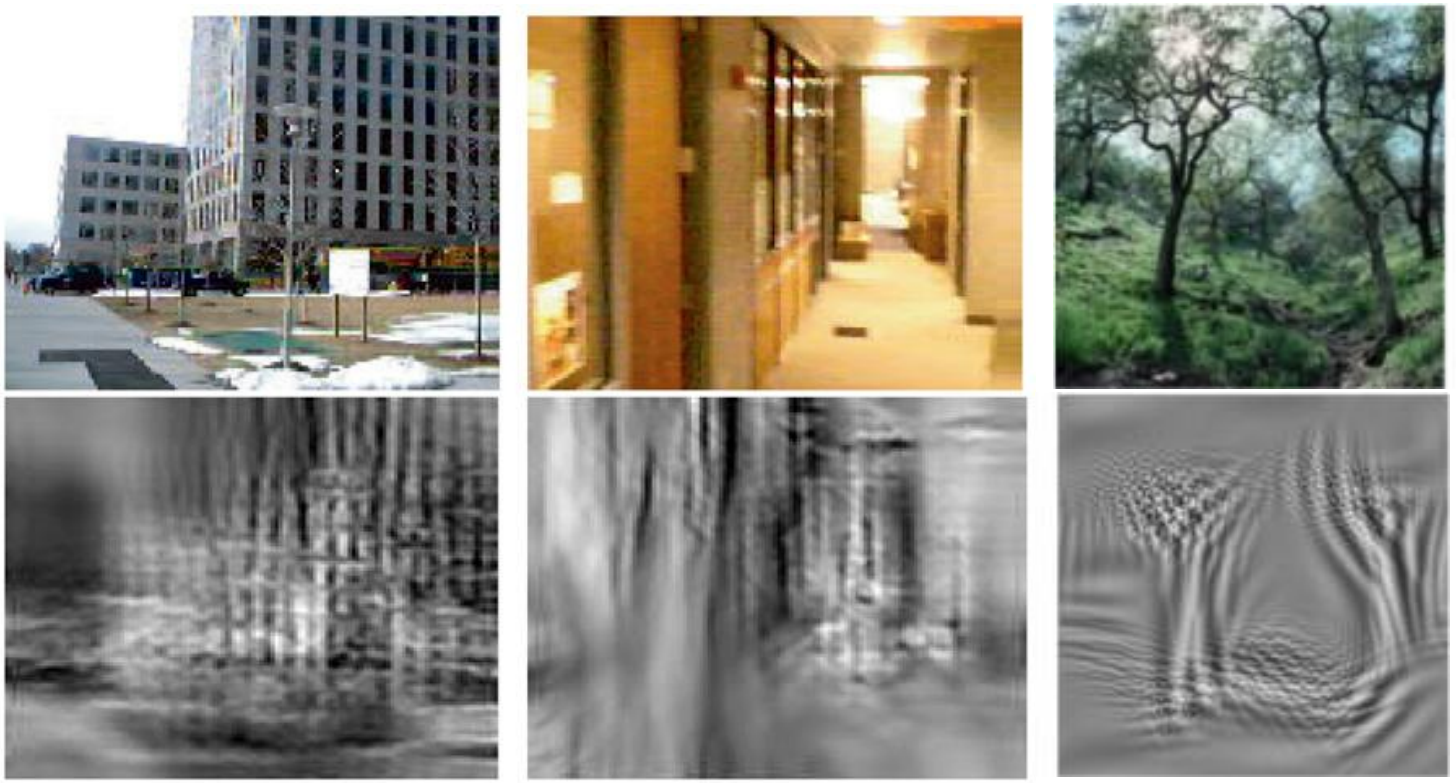
- No distinction between foreground and background: scene recognition?

Timeline of recognition

- 1965-late 1980s: alignment, geometric primitives
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- Early 2000s – present : parts-and-shape models
- 2003 – present: bags of features
- Present trends: combination of local and global methods, modeling context, integrating recognition and segmentation

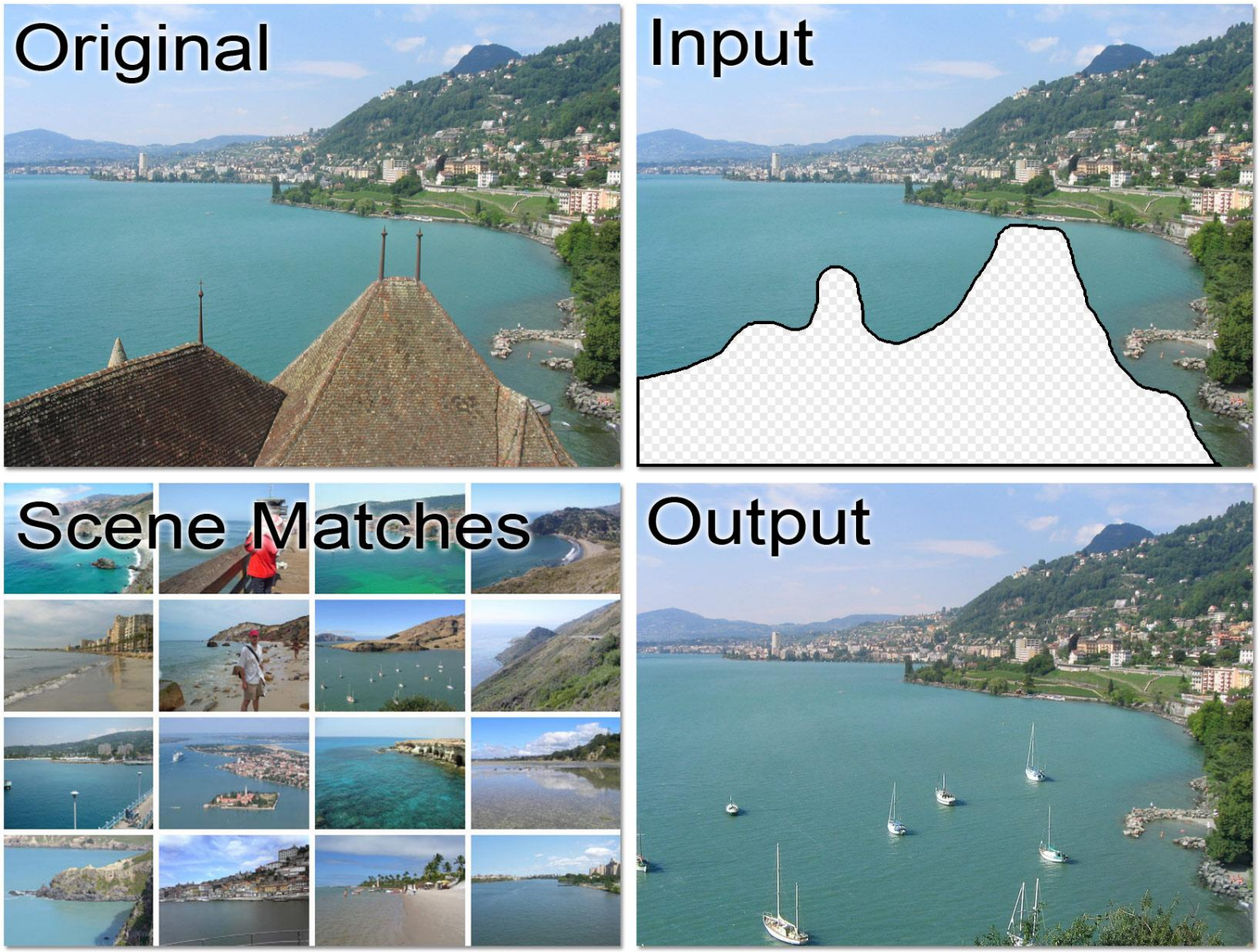
Global models?

- The “gist” of a scene: Oliva & Torralba (2001)



Oliva & Torralba, Building the gist of a scene the role of global image features in recognition, 2001

J. Hays and A. Efros, Scene Completion using Millions of Photographs, SIGGRAPH 2007



Object Recognition by Scene Alignment

Bryan C. Russell, Antonio Torralba, Ce Liu, Rob Fergus, William T. Freeman

NIPS 2007



Input image

Goal: Recognize objects embedded in a scene

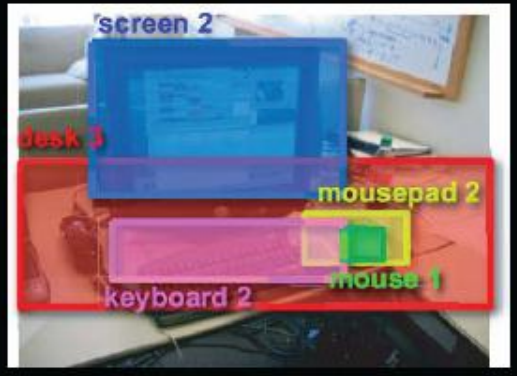
Object Recognition by Scene Alignment



Nearest neighbors from 15,691 images



Cluster images using object labels



Output image with object labels transferred

The end !